*A project report on*

# VERIDEX: LLM-DRIVEN ADVERSARIAL EVALUATION OF MANIPULATION DETECTION IN SIMULATED COMMODITY FUTURES MARKETS

*Submitted in partial fulfillment for the award of the degree of*

**Bachelor of Technology in PROGRAM NAME WITH SPECIALIZATION**

*by*

**YOUR NAME (REG. NO.)**




**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**

November, 2026

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I hereby declare that the thesis entitled "Veridex: LLM-Driven Adversarial Evaluation of Manipulation Detection in Simulated Commodity Futures Markets" submitted by YOUR NAME (REG. NO.), for the award of the degree of Bachelor of Technology in PROGRAM NAME WITH SPECIALIZATION, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision of GUIDE NAME.

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PROGRAM NAME WITH SPECIALIZATION

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# ABSTRACT

Spoofing, layering, wash trading and pump-and-dump schemes distort commodity futures prices. Surveillance systems are usually evaluated on historical cases, which says little about how they hold up once manipulators adapt. Veridex is a fully simulated surveillance laboratory built to study that question.

Veridex has five modules. An agent-based limit order book simulator, calibrated to eight commodity futures, generates realistic markets with noise, momentum, mean-reversion and market-making traders, high-frequency liquidity providers and institutional execution algorithms, together with four manipulation personas whose ground truth is logged exactly. A two-layer detector computes 51 trader-level features over rolling three-minute windows and combines an unsupervised Isolation Forest with four persona-specific XGBoost classifiers, explaining every alert with SHAP values. An adversarial evaluation loop then lets a large language model strategist (Cerebras gpt-oss-120b) re-tune the manipulators each round, scores the mutated schemes against the frozen detector before retraining, and penalises any mutation that stops manipulating. An LLM layer also writes grounded case notes and answers analyst questions, and a Streamlit dashboard presents the results.

In a six-round experiment over eight commodities (about 12.8 million order events and 2.6 million trades per round), the baseline detector reached F1 scores of 0.98 for spoofing, 0.97 for wash trading, 0.99 for layering and 0.72 for pump-and-dump. The LLM strategist's mutations reduced the frozen detector's coverage to 0.14 for pump-and-dump and 0.62 for wash trading in round 1, and to 0.59 for spoofing and 0.69 for layering in round 2, while keeping at least 62 percent of each scheme's original market effect. Cumulative retraining recovered most of the loss, with post-retrain F1 scores between 0.71 and 0.87 in the final round. The run cost about USD 0.07 in API credit. Adversarial, LLM-driven evaluation thus exposes weaknesses that static evaluation hides, which regular retraining largely repairs.



**Keywords:** market manipulation, spoofing, agent-based simulation, limit order book, anomaly detection, XGBoost, SHAP, adversarial machine learning, large language model agents.

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Place: Chennai

Date:

**YOUR NAME**

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# LIST OF ACRONYMS

| | |
|---|---|
| **ABM** | Agent-Based Model |
| **API** | Application Programming Interface |
| **AUC** | Area Under the Curve |
| **CFTC** | Commodity Futures Trading Commission (United States) |
| **CI** | Confidence Interval |
| **CV** | Cross-Validation / Coefficient of Variation |
| **DOJ** | Department of Justice (United States) |
| **EMA** | Exponential Moving Average |
| **F1** | Harmonic mean of precision and recall |
| **FPR** | False Positive Rate |
| **HFT** | High-Frequency Trading |
| **IF** | Isolation Forest |
| **LLM** | Large Language Model |
| **LOB** | Limit Order Book |
| **MAR** | Market Abuse Regulation (European Union) |
| **ML** | Machine Learning |
| **OHLCV** | Open, High, Low, Close, Volume |
| **OOF** | Out-Of-Fold |
| **OTR** | Order-to-Trade Ratio |
| **OU** | Ornstein–Uhlenbeck (mean-reverting stochastic process) |
| **P&L** | Profit and Loss |
| **SHAP** | SHapley Additive exPlanations |
| **SRS** | Software Requirements Specification |
| **TTL** | Time To Live (of a resting order) |
| **UI** | User Interface |
| **XGBoost** | eXtreme Gradient Boosting |

# Chapter 1
# Introduction

## 1.1 BACKGROUND

Commodity futures markets set the reference prices for crude oil, natural gas, metals and grains. Refiners, airlines, miners, millers and farmers use these prices to plan production and hedge risk, and the same contracts are traded by banks, proprietary trading firms, asset managers and retail participants. Almost all of this trading now happens on electronic exchanges that match orders in a central limit order book (LOB). The book is visible to every participant: anyone can see how much is offered to buy and to sell at each price level, and most trading algorithms react to that information within milliseconds.

That visibility is also what makes manipulation possible. A trader who places large orders that they never intend to execute can make the book look heavier on one side, so other participants move their prices; the manipulator then trades at the improved price and cancels the fake orders. This is *spoofing*. Placing several such orders at consecutive price levels is *layering*. Trading with oneself or with colluding accounts creates the appearance of volume without any change in ownership, which is *wash trading*. Buying aggressively to trigger momentum traders and then selling into the demand that follows is a *pump-and-dump* or momentum-ignition scheme. All four are prohibited: in the United States, Section 747 of the Dodd-Frank Wall Street Reform and Consumer Protection Act (2010) made spoofing explicitly illegal, and the European Union's Market Abuse Regulation (Regulation (EU) No 596/2014, 2014) prohibits giving false or misleading signals about supply, demand or price.

Enforcement cases show that these schemes are not hypothetical. Navinder Sarao's spoofing of E-mini S&P 500 futures was linked by regulators to the 2010 “flash crash”; Michael Coscia became the first trader criminally convicted under the Dodd-Frank anti-spoofing provision; JPMorgan Chase agreed in 2020 to pay about USD 920 million to resolve spoofing in precious-metals and Treasury futures; and Tower Research Capital paid USD 67.4 million in 2019 for layering by its traders. Each of these schemes ran for months or years before it was stopped, which raises an obvious question for anyone building surveillance software: how well does a detector keep working once the people it is meant to catch start adapting to it?

## 1.2 MOTIVATION

Most published work on manipulation detection trains a model on labelled historical data, or on a fixed synthetic data set, and reports precision and recall on a held-out portion of the same data. This measures how well a detector recognises the manipulation it has already seen. It does not measure what happens when a manipulator changes tactics, for example by splitting a large spoof order into smaller pieces, randomising cancel timing, spreading activity across several accounts, or trading only during busy periods. Real manipulators did exactly these things; the enforcement records describe them in detail.

Testing a detector against adaptive manipulators is hard for three reasons. First, real labelled manipulation data is scarce, confidential and legally sensitive, and it contains only the schemes that were caught. Second, producing new evasive variants by hand is slow and limited by the imagination of the person writing them. Third, an “evasion” that simply stops manipulating (for example, a spoof so small it no longer moves the market) is not a real evasion, so any evaluation must check that the mutated scheme still works.

Two recent developments make it possible to address all three. Agent-based market simulation can generate realistic order-book data in which the ground truth is known exactly, because the simulator itself authors every manipulation episode. And large language models (LLMs) can now reason over structured feedback, read a detector's explanations and call tools, which makes them suitable for playing the role of an adaptive adversary that proposes new evasion tactics each round. Veridex combines these ideas into a closed-loop red-team environment for market surveillance.

## 1.3 PROBLEM STATEMENT

The problem addressed by this project can be stated as follows. Given a simulated commodity futures market with known manipulation ground truth, build a manipulation detector that identifies the manipulating accounts and explains its decisions, and then measure how its detection performance changes when an automated adversary repeatedly re-tunes the manipulation strategies to evade it, under the constraint that every mutated strategy must still achieve a meaningful fraction of its original market effect.

The detector must work at the level of individual trader accounts in short time windows, because that is how surveillance analysts investigate cases. It must distinguish manipulators from legitimate participants whose behaviour superficially resembles manipulation, such as high-frequency market makers that cancel most of their orders and institutional execution algorithms that rest and withdraw large orders. The adversary must be bounded: it can change only documented strategy parameters within validated limits, and it must not be able to "win" by doing nothing. Finally, the whole system must run on a student budget: the LLM is accessed through a paid API with about USD 5 of credit, so every call must be justified, cached and capped.

## 1.4 OBJECTIVES

The objectives of the project are:

1. To build a reproducible agent-based limit order book simulator, calibrated to eight commodity futures, in which legitimate traders and four manipulation personas (spoofing, layering, wash trading and pump-and-dump) interact, and every manipulation episode is logged as ground truth.
2. To design trader-level behavioural features and a two-layer detector, combining an unsupervised Isolation Forest with persona-specific supervised classifiers, that flags manipulating accounts and explains each alert with SHAP values.
3. To build an adversarial evaluation loop that measures detection both before retraining (against a frozen detector) and after retraining, together with a measure of each manipulation's market effect, so that evasion and effectiveness are judged together.
4. To use an LLM as an adaptive adversarial strategist that reads the detector's performance and explanations and proposes new manipulation parameters each round, with code-level guards that keep its proposals valid and effective.
5. To use the same LLM layer to generate grounded case notes for flagged accounts and to answer analyst questions about a run, while keeping API cost within a fixed budget.
6. To present the markets, alerts, explanations, model performance and adversarial history in an interactive dashboard suitable for analysts and examiners.

## 1.5 SCOPE OF THE PROJECT

Everything in Veridex is simulated. The project does not connect to any exchange, does not trade, does not use real accounts and does not produce real regulatory findings; every dashboard page carries a notice to that effect. Real market data is used only to calibrate the simulator (volatility, drift, tick size and typical volume of each commodity). The four manipulation personas are modelled on publicly documented enforcement cases. Cross-market manipulation (for example, spoofing one contract to move a related one), cornering and “banging the close” are outside the current scope. The detector is evaluated on simulated test markets that are generated independently of the training markets in every round.

## 1.6 CONTRIBUTIONS

The main contributions of this work are:

- **A realistic, adversarially useful market simulator.** Beyond the usual noise, momentum and mean-reversion agents, it includes high-frequency liquidity providers and institutional execution algorithms whose behaviour overlaps with manipulation, so that the detection problem is not trivially separable.
- **An evaluation protocol that judges evasion and effectiveness together.** Each mutation is scored on how much of each episode the frozen detector missed and how late it was caught, multiplied by how much of the original market effect the scheme kept. Being flagged under another persona's label still counts as being caught.
- **A bounded, cost-controlled LLM adversary** that uses tool calls, a structured track record, the detector's SHAP reliance and “tactic cards” derived from enforcement cases, protected by an automatic effect guard and a hard budget ledger.
- **An explainable analyst workflow**: SHAP explanations, a confidence panel, an evidence table, LLM case notes checked against the evidence, and a tool-grounded chat assistant, all presented in a Streamlit dashboard.

## 1.7 ORGANISATION OF THE REPORT

The rest of this report is organised as follows. Chapter 2 reviews the literature on market manipulation, its enforcement, market simulation, detection methods, explainability, adversarial machine learning and LLM agents, and identifies the research gap. Chapter 3 lists the requirements and the development environment. Chapter 4 presents the overall architecture and design. Chapters 5, 6 and 7 describe the implementation of each module: the market simulator; the detection engine and adversarial evaluation loop; and the LLM layer and dashboard. Chapter 8 presents and discusses the results of the six-round experiment. Chapter 9 concludes and outlines future work. References and appendices containing configuration listings, parameter bounds and sample LLM exchanges follow.

# Chapter 2
# Literature Survey

## 2.1 MARKET MANIPULATION IN FUTURES MARKETS

Economic theory distinguishes three broad kinds of manipulation (Allen & Gale, 1992): *action-based* manipulation, which changes the actual value of an asset; *information-based* manipulation, which spreads false information; and *trade-based* manipulation, in which the manipulator profits purely through trading, without any public action or false statement. All four schemes studied in this project are trade-based. Allen and Gale showed that trade-based manipulation can be profitable even with rational market participants, because uninformed traders cannot be sure whether a large buyer is informed or manipulating. Aggarwal and Wu (2006) studied enforcement cases from the U.S. Securities and Exchange Commission and found that manipulation is more common in less liquid markets and that manipulators tend to trade during periods of elevated volume, a tactic modelled in Veridex as the "busy period" knob. Putniņš (2012) surveyed the field and noted that empirical work is limited by the fact that only detected manipulation is ever observed.

Spoofing has received particular attention since high-frequency trading became dominant. Lee, Eom and Park (2013) analysed spoofing traders in the Korean stock market and showed that they place large orders away from the best price and cancel them quickly, and that the strategy is profitable. Cartea, Jaimungal and Wang (2020) modelled spoofing in an order-driven market and showed that because other traders use order-book imbalance to predict short-term price moves, a trader who can distort imbalance with orders that are unlikely to execute can profit. This mechanism (fake depth shifts the imbalance, the imbalance shifts other participants' quotes) is exactly how spoofing and layering are implemented in the Veridex simulator. Wash trading has been studied extensively in cryptocurrency exchanges; Cong, Li, Tang and Yang (2023) estimated that a large share of reported volume on unregulated exchanges was wash traded. Pump-and-dump schemes in thin markets were characterised by Kamps and Kleinberg (2018), who proposed detection rules based on sudden price and volume increases.

## 2.2 REGULATION AND ENFORCEMENT CASES

Regulators enforce anti-manipulation rules through surveillance and investigation. Table 2.1 summarises the enforcement cases that shaped the four Veridex personas and the evasion tactics available to the LLM strategist (Commodity Futures Trading Commission [CFTC], 2015, 2019, 2020, 2021; CFTC v. Kraft Foods Group, 2015; CFTC v. Oystacher, 2015; United States v. Coscia, 2017). The persona parameters and tactic cards cite these cases directly, so that every evasion the strategist can try corresponds to behaviour that real manipulators actually used.

**Table 2.1: Enforcement cases used to design the manipulation personas**

| Case | Scheme | Documented tactics | Use in Veridex |
|---|---|---|---|
| Sarao / Nav Sarao Futures (CFTC, 2015) | Spoofing / layering in E-mini S&P 500 futures | Large sell orders kept a few levels from the best price and continually re-priced as the market moved | Spoofing knob `reprice_to_touch` |
| Coscia (U.S. v. Coscia, convicted 2015, affirmed 2017) | Spoofing in energy and metals futures | Large orders cancelled within fractions of a second after a small opposite order filled; later split into batches | Baseline spoofing cycle; split orders; cancel-timing jitter |
| Oystacher / 3Red Trading (CFTC, 2015) | Spoofing with "flipping" | After pulling spoof orders, trading aggressively in the other direction | Spoofing knob `flip_after_cancel` |
| JPMorgan Chase (CFTC/DOJ, 2020, about USD 920 million) | Spoofing in precious metals and Treasuries | Genuine and spoof orders entered by different traders on the same desk | Spoofing knob `n_accounts` (cross-account legs) |
| Tower Research Capital (CFTC/DOJ, 2019, USD 67.4 million) | Layering | Several orders on consecutive price levels, cancelled together after the opposite order filled | Layering persona: layer count, spacing, size taper |
| Kraft Foods / Mondelez (CFTC, complaint 2015, settled 2019) | Wheat futures manipulation including wash trades | Related accounts trading with each other while staying flat | Wash trading persona; camouflage trades |
| Coinbase (CFTC, 2021, USD 6.5 million) | Wash trading by automated programs | Two internal programs repeatedly matching each other's orders | Wash rings across several accounts |

Regulators have also published the patterns they look for. Staff guidance from the U.S. Commodity Futures Trading Commission (CFTC) highlights large order sizes relative to the market, short order lifetimes, high cancellation rates, orders on the opposite side of executions, and activity concentrated around busy periods. These observations are encoded both in the detector's features (Chapter 6) and, from the attacker's side, in the tactic cards given to the LLM strategist (Chapter 7).

## 2.3 AGENT-BASED SIMULATION OF FINANCIAL MARKETS

Agent-based models (ABMs) simulate a market as a population of interacting traders with simple behavioural rules. Early models such as the Santa Fe artificial stock market

showed that realistic price dynamics can emerge from heterogeneous agents. Chiarella and Iori (2002) built an order-driven market with fundamentalist, chartist and noise traders and showed that the mix of strategies reproduces fat-tailed returns and volatility clustering. Gould et al. (2013) reviewed the empirical properties and models of limit order books, including price-time priority matching, which Veridex implements exactly. Byrd, Hybinette and Balch (2020) introduced ABIDES, a discrete-event multi-agent simulator for market research. Avellaneda and Stoikov (2008) derived optimal quoting for a market maker that skews its quotes according to its inventory; the Veridex market maker follows this idea.

A credible simulator must reproduce the *stylized facts* of real returns (Cont, 2001): heavy tails (positive excess kurtosis), volatility clustering (positive autocorrelation of absolute returns) and little linear predictability of returns themselves. Veridex checks these properties for every commodity (Section 5.7). The fundamental value of each commodity follows an Ornstein–Uhlenbeck mean-reverting process (Uhlenbeck & Ornstein, 1930), a standard model for commodity prices because of storage and supply responses (Deaton & Laroque, 1996; Geman, 2005), with parameters calibrated from published historical statistics (Erb & Harvey, 2006).

## 2.4 APPROACHES TO MANIPULATION DETECTION

Detection approaches fall into three groups. *Rule-based surveillance*, still common in industry, raises an alert when a hand-set threshold is crossed, for example an order-to-trade ratio above a limit. Rules are transparent but easy to evade once the thresholds are known, and they generate many false alarms on legitimate high-frequency traders. *Statistical and unsupervised methods* model normal behaviour and flag deviations; they need no labels, which suits a domain where labels are scarce. The Isolation Forest (Liu, Ting & Zhou, 2008) isolates anomalies with random partitioning trees: anomalous points are separated in fewer splits, so the average path length gives an anomaly score. It scales well and works with mixed feature types. *Supervised methods* learn from labelled examples and are usually more accurate on known schemes. Gradient-boosted decision trees, and XGBoost in particular (Chen & Guestrin, 2016), perform strongly on tabular data, handle class imbalance through instance weighting and are efficient enough to retrain every round.

**Table 2.2: Comparison of manipulation detection approaches**

| Approach | Strengths | Weaknesses | Role in Veridex |
|---|---|---|---|
| Threshold rules | Transparent, simple, auditable | Easy to evade; many false positives on HFT | Basis for intuition, not used directly |
| Unsupervised anomaly detection (Isolation Forest) | No labels needed; catches unknown schemes | Flags any unusual account, including legitimate outliers | Trader-level and market-level forests |
| Supervised classification (XGBoost) | High accuracy on known schemes; fast to retrain | Needs labels; may miss novel variants | One classifier per persona |
| Graph / network analysis | Natural fit for wash rings and collusion | Needs counterparty graphs; costly | Approximated by counterparty features |
| Adversarial evaluation (this work) | Measures robustness to adaptive manipulators | Needs a simulator and an adversary | Core of Modules 3 and 4 |

## 2.5 EXPLAINABLE MACHINE LEARNING

A surveillance alert is only useful if an analyst can understand why it was raised, and regulators expect firms to be able to justify their alerts. SHAP (SHapley Additive exPlanations) attributes a model's prediction to its input features using Shapley values from cooperative game theory (Lundberg & Lee, 2017). For tree ensembles, TreeSHAP computes exact Shapley values in polynomial time (Lundberg et al., 2020). Veridex stores the five most important SHAP contributions with every alert, shows them in the dashboard, and passes the detector's mean absolute SHAP values to the adversarial strategist so it knows which features the detector currently relies on.

## 2.6 ADVERSARIAL MACHINE LEARNING AND RED TEAMING

Adversarial machine learning studies how models fail under deliberate manipulation of their inputs. Goodfellow, Shlens and Szegedy (2015) showed that small, targeted perturbations can fool neural networks, and Biggio and Roli (2018) reviewed a decade of evasion and poisoning attacks, emphasising that security evaluation should assume an adaptive attacker who knows how the defence works. In fraud and surveillance settings the attacker is constrained: an evasion must still achieve the attacker's goal, so the realistic threat model is “evade while still manipulating”. This project encodes that constraint directly in the adversary's objective (Section 6.9). The loop of attack, measurement and retraining also resembles red-team exercises, in which a dedicated team tries to defeat a system so that its weaknesses can be fixed before real attackers find them.

## 2.7 LARGE LANGUAGE MODEL AGENTS AND TOOL USE

Large language models can follow instructions, reason over structured context and call external functions. ReAct (Yao et al., 2023) interleaves reasoning with actions, and Toolformer (Schick et al., 2023) showed that models can learn when to call tools. Agentic use of LLMs, in which the model selects and sequences tool calls to complete a bounded task, is now common. Open-weight models such as gpt-oss-120b (OpenAI, 2025) can be served at high speed by inference providers such as Cerebras, which makes them practical for short, structured decisions. Veridex uses an LLM in three bounded roles: an adversarial strategist that calls a single validated `propose_mutation` tool, a narrative generator whose output is checked against the evidence it was given, and an analyst assistant that answers only from data returned by read-only query tools.

## 2.8 RESEARCH GAP

The literature provides the individual building blocks: realistic order-book simulation, unsupervised and supervised detection, explanation methods, adversarial evaluation and LLM agents. What is missing is an end-to-end, reproducible environment that puts them together to answer a practical question: *does a manipulation detector stay effective when manipulators adapt to it, and what does an adaptive manipulator learn to do?* Existing detection studies evaluate on static data; existing simulators are rarely paired with explainable detectors; and adversarial studies rarely enforce that an evasion must still manipulate. Veridex addresses this gap with a closed loop in which a bounded LLM adversary mutates documented manipulation tactics, a detector is measured before and after retraining, and every mutation's market effect is checked against the baseline.

# Chapter 3
# Requirements and Analysis

This chapter summarises what Veridex must do and the conditions it must satisfy. The requirements were written before implementation in a Software Requirements Specification (SRS) with stable identifiers (FR-x.y for functional requirements, NFR-x for non-functional ones), and the implementation was checked against them. The status column reports honestly what the final system delivers.

## 3.1 STAKEHOLDERS AND USERS

Three kinds of user were considered. A *surveillance analyst* reviews flagged accounts, reads explanations and asks questions about a run. A *researcher* configures simulations, runs the adversarial loop and studies how detection changes across rounds. An *examiner* (for example, a project evaluator) needs to understand each screen quickly without verbal explanation. All three use the same dashboard; researchers additionally use the command-line pipeline.

## 3.2 FUNCTIONAL REQUIREMENTS

**Table 3.1: Functional requirements and implementation status**

| ID | Requirement (summary) | Status |
|---|---|---|
| FR-0.1 | Modules communicate only through schema-validated files on disk (Parquet / YAML / JSON) | Implemented |
| FR-0.2 | Every data file is validated against its schema before it is written | Implemented (pandera) |
| FR-0.3 | Each run has a unique run_id directory with configuration, seed and timestamp | Implemented |
| FR-0.4 | Same seed and configuration give identical output logs | Implemented; tested |
| FR-1.1 | Limit order book with price-time priority, no learned component in price formation | Implemented |
| FR-1.2 | At least four legitimate agent archetypes sharing one decide() interface | Implemented (six types) |
| FR-1.3 | At least three manipulation personas trading in the same book, independently tunable | Implemented (four) |
| FR-1.4 | Layering (should) and cornering (could) as extensions | Layering implemented; cornering not |
| FR-1.5 | Every trade and every order lifecycle event logged | Implemented |
| FR-1.6 | Ground-truth manipulation windows logged with accounts and parameters | Implemented |
| FR-1.7 | Trades resampled to OHLCV bars | Implemented |
| FR-1.8 | Calibration targets estimated from real commodity data, with fallback constants | Implemented |
| FR-1.9 | Replay from any tick with a modified population (counterfactual) | Not implemented |
| FR-2.1 | Rolling-window, per-trader behavioural features | Implemented (51 features) |
| FR-2.2 | Unsupervised anomaly model trained without labels | Implemented (Isolation Forest) |

| ID | Requirement (summary) | Status |
|---|---|---|
| FR-2.3 | One supervised classifier per persona | Implemented (XGBoost) |
| FR-2.4 | SHAP attributions for supervised predictions | Implemented |
| FR-2.5 | Single ensemble flag per (window, trader) with triggering layers and top features | Implemented |
| FR-2.6 | Precision, recall, F1 and FPR reported per persona | Implemented |
| FR-2.7 | Detection latency per detected episode | Implemented |
| FR-3.1 | Round 0 baseline with obvious personas | Implemented |
| FR-3.2 | Each round: new data with mutated personas, pre-retrain and post-retrain evaluation | Implemented |
| FR-3.3 | Hand-authored mutation library grounded in enforcement cases | Implemented (fallback) |
| FR-3.4 | SHAP feature-importance drift between rounds | Implemented |
| FR-3.5 | Price impact avoided by early detection (counterfactual replay) | Not implemented |
| FR-4.1 | Case narratives derived only from computed evidence | Implemented with grounding check |
| FR-4.2 | Bounded tool-using agent that proposes next-round mutations | Implemented |
| FR-4.3 | Analyst assistant answering only from logged data through function calls | Implemented |
| FR-4.4 | Graceful degradation when the LLM is unavailable | Implemented |
| FR-4.5 | Cached LLM outputs so a demonstration can run without new API calls | Implemented (response cache) |
| FR-4.6 | Every LLM call logged with tokens, latency and cost | Implemented |
| FR-5.1 to 5.6 | Dashboard: market charts with manipulation shading, performance curves, case review, chat, run selection | Implemented (5.5 omitted) |

## 3.3 NON-FUNCTIONAL REQUIREMENTS

**Table 3.2: Key non-functional requirements**

| Category | Requirement | How it is met |
|---|---|---|
| Performance | A 10,000-tick market with the full population completes in minutes on a laptop, without a GPU | About 15–20 s per commodity; the 8 commodities run in parallel processes |
| Reliability | An LLM failure never breaks the simulation or the detector | Fallback mutation library; LLM features are disabled with an explanation |
| Cost | Total LLM spending stays within about USD 5 of API credit | Hard budget ledger capped at USD 4.50, response cache, call spacing, per-process call cap |
| Security | API keys never committed; prompts separate instructions from data | .env is git-ignored; keys stripped from logs; tool output treated as data |
| Reproducibility | Identical seed and configuration give identical output | Seeded generators per agent; deterministic order ids; reproducibility test |
| Maintainability | Edge cases covered by automated tests; schemas defined once | 118 automated tests; all schemas in src/utils/schemas.py |
| Usability | A first-time viewer understands each screen quickly | Titles, captions, hover help, colour plus icon for each persona |
| Ethics | Never presented as a real surveillance product; no real trader data | Synthetic ids only; “simulation output” notice on every page |

## 3.4 USE CASES

Five use cases drove the design. They are summarised below; Chapter 7 shows the screens that implement them.

**Table 3.3: Principal use cases**

| ID | Use case | Actor | Main flow |
|---|---|---|---|
| UC-1 | Run a full adversarial experiment | Researcher | Start the pipeline with rounds, ticks and seed; the system simulates, trains, scores and mutates round by round |
| UC-2 | Review a flagged case | Analyst | Select a round and market, open a flagged account, read confidence, SHAP and evidence, generate a case note |
| UC-3 | Ask a question about a run | Analyst | Type a question in the chat; the assistant calls query tools and answers with cited sources |
| UC-4 | Propose the next mutation | LLM strategist | Read track record, SHAP reliance and tactic cards; call propose_mutation for each persona |
| UC-5 | Inspect round-by-round performance | Examiner | Open Model Performance; compare frozen and retrained detection, effect and strategist decisions |

## 3.5 HARDWARE AND SOFTWARE ENVIRONMENT

**Table 3.4: Development and execution environment**

| Item | Specification |
|---|---|
| Operating system | Windows 11 Pro (64-bit); the code is platform-independent Python |
| Processor / memory | Standard multi-core laptop CPU; no GPU required |
| Language | Python 3.14 in a project virtual environment |
| Data and numerics | pandas 2.3, NumPy 2.5, SciPy 1.17, PyArrow 25 (Parquet) |
| Validation | pandera 0.33 (DataFrame schemas) |
| Machine learning | scikit-learn 1.9 (Isolation Forest, GroupKFold), XGBoost 3.4, SHAP 0.49 |
| LLM access | LiteLLM 1.100; Cerebras gpt-oss-120b (primary) or Google Gemini (alternative) |
| Dashboard | Streamlit 1.63, Plotly 7.0 |
| Testing | pytest 9.1, Hypothesis (property-based tests) |
| Configuration | YAML files under configs/; secrets in a git-ignored .env file |
| Version control | Git, with the project hosted on GitHub |

## 3.6 CONSTRAINTS AND ASSUMPTIONS

- The market is simulated; real data is used only for calibration constants. One simulation tick represents one second of trading, and a trading day is 23,400 ticks (6.5 hours).
- Ground truth is exact: an account-window is labelled as manipulation only if the account belongs to an episode and the window overlaps the episode by at least 15 ticks.
- The LLM is an untrusted component. Its proposals are validated against hard parameter bounds, its outputs are logged, and every function it can call is either read-only or validated.
- API credit is limited, so every call must have a clear purpose and identical requests must be answered from cache.

# Chapter 4
# System Design

## 4.1 ARCHITECTURAL OVERVIEW

Veridex is organised as five modules around a shared configuration layer (Figure 4.1). Module 1, the agent-based market simulator, produces order, trade, bar and ground-truth logs. Module 2, the detection engine, turns those logs into trader-level features and scores every account in every window. Module 3, the adversarial evaluation loop, runs the experiment round by round: it simulates new markets, scores them with the frozen detector, retrains, scores again and records how effective each manipulation remained. Module 4, the LLM and agentic layer, reads Module 3's results to propose the next round's manipulation parameters, and also writes case notes and answers analyst questions. Module 5, the dashboard, presents all of these artefacts.

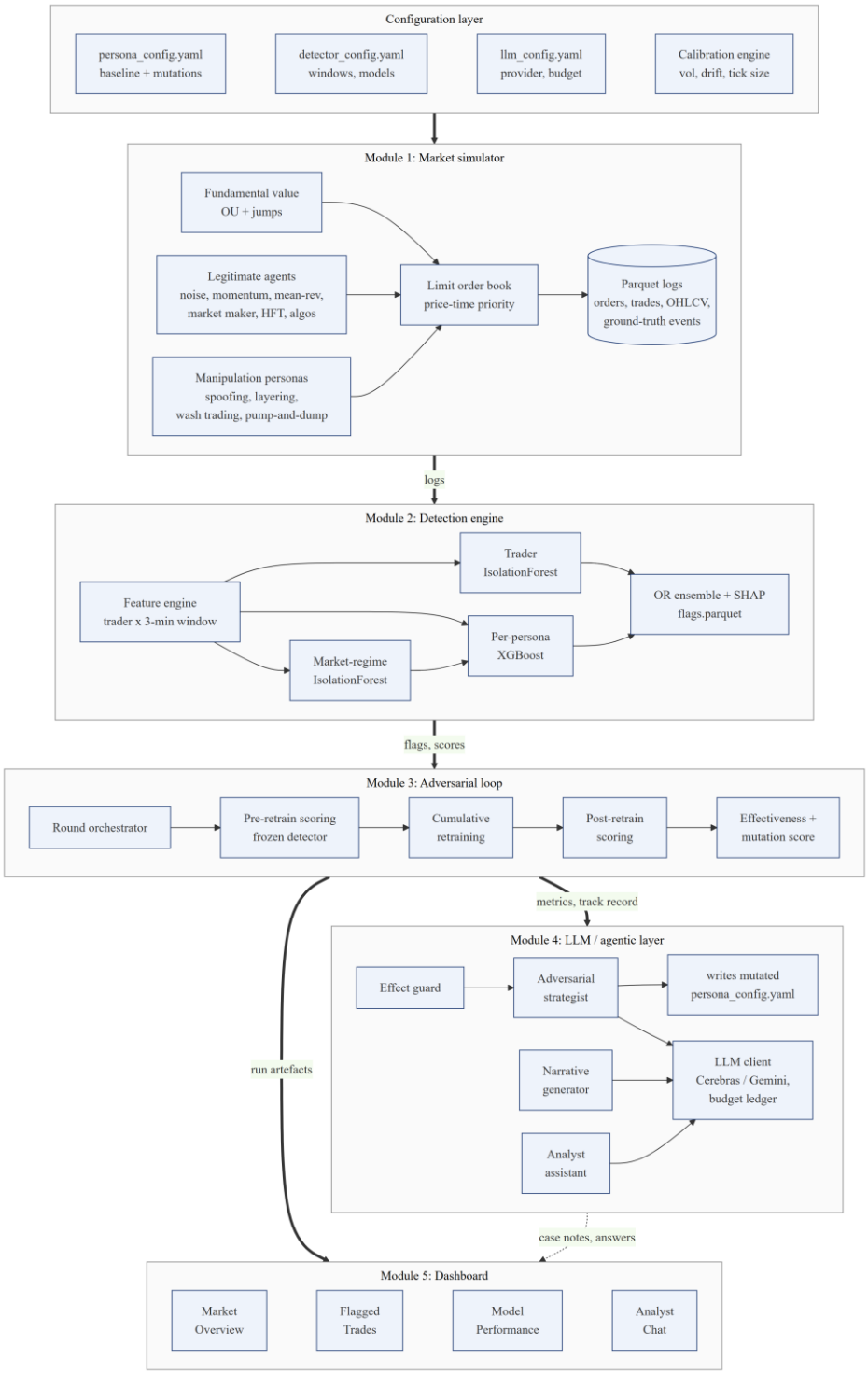



**Figure 4.1: Module-wise architecture of Veridex (rendered from a Mermaid diagram)**

Two design principles run through the architecture. First, modules communicate only through files on disk, each validated against a declared schema before it is written (SRS FR-0.1 and FR-0.2). This keeps the modules independent: the dashboard, for instance, can display any completed run without importing the simulator, and a run can be

analysed long after it was produced. Second, every source of randomness is seeded, so a run can be reproduced exactly from its seed and configuration.

## 4.2 MODULE DECOMPOSITION

**Table 4.1: Modules and their source packages**

| Module | Package / files | Responsibility |
|---|---|---|
| Configuration | configs/*.yaml, src/core/calibration.py, src/core/persona_params.py | Calibration constants, persona parameters (baseline and mutations), detector and LLM settings |
| 1. Market simulator | src/core/order_book.py, simulation.py, agents/, *agents/manipulation/* | Fundamental value, order book, legitimate agents, manipulation personas, logging |
| 2. Detection engine | src/detector/features.py, anomaly.py, classifiers.py, pipeline.py | Features, Isolation Forests, XGBoost classifiers, ensemble, SHAP, flags |
| 3. Adversarial loop | src/adversarial/orchestrator.py, metrics.py, effectiveness.py; scripts/run_full_pipeline.py | Round orchestration, pre/post metrics, episode coverage, effectiveness, mutation score |
| 4. LLM layer | src/agentic/llm_client.py, adversarial_strategist.py, tools.py, mutation_library.py, narrative_generator.py, case_evidence.py, analyst_assistant.py | Provider switch and budget, strategist, effect guard, case notes, analyst chat |
| 5. Dashboard | dashboard/app.py, data_loader.py, components/* | Market Overview, Flagged Trades, Model Performance and Analyst Chat tabs |
| Shared utilities | src/utils/schemas.py, io.py, logging_config.py | Schemas, validated I/O, logging |

In total the project contains about 15,400 lines of Python across the source, dashboard, scripts and tests.

## 4.3 DATA DESIGN

Every simulated market writes four Parquet files, and every evaluated round adds detector outputs. Table 4.2 lists the main data contracts; each is defined once as a pandera schema and validated on write.

**Table 4.2: Core data schemas**

| Artefact | Key fields | Produced by / used by |
|---|---|---|
| order_log | order_id, commodity, trader_id, side, price, quantity, tick, event_type (placed, cancelled, filled, partial_fill, rejected_cancel) | Simulator / feature engine, dashboard |
| trade_log | trade_id, commodity, tick, price, quantity, buyer_trader_id, seller_trader_id, aggressor_side | Simulator / feature engine, effectiveness, dashboard |
| ohlcv_bars | bar_start, commodity, open, high, low, close, volume | Simulator / dashboard |
| manipulation_events | event_id, commodity, persona, start_tick, end_tick, trader_ids, parameters | Simulator / labels, effectiveness, evaluation |
| features / labels | (commodity, window_start, trader_id) index; 51 features; one 0/1 column per persona | Feature engine / detector training and scoring |
| flags | final_flag, final_score, iso_score, predicted_persona, triggered_by, top_shap_features, persona_scores | Detector / dashboard, narrative generator, analyst tools |
| metrics.json | Per persona: post- and pre-retrain precision, recall, F1, FPR, coverage, latency; effectiveness, ratios, scores, parameters | Adversarial loop / strategist, dashboard |

| Artefact | Key fields | Produced by / used by |
|---|---|---|
| strategist.json | Per persona: source (LLM or hand-authored), parameter changes, reasoning, guard actions | Pipeline / dashboard |

A run is stored as `data/runs/<run_id>/round_<N>/`, with a `train` market and two test markets (`test` and `test_1`) per round, each containing the logs for all eight commodities, plus the round's metrics, flags, thresholds and SHAP importance. The LLM audit log `llm_calls.jsonl` and the strategist's reasoning log sit at the run level.

## 4.4 ADVERSARIAL ROUND DESIGN

Figure 4.2 shows the control flow of one experiment. Round 0 establishes the baseline: the personas use their documented, obvious parameters, the detector is trained on the round-0 training market, and the test markets give the baseline metrics and the baseline market effect of each scheme. Every later round has a mutation step and an evaluation step. In the mutation step the effect guard first checks whether any persona's effect collapsed in the previous round; the strategist then proposes new parameters, which are validated against hard bounds, and the at-risk rule is enforced. In the evaluation step new markets are simulated with the mutated personas and scored twice: first by the frozen detector from the previous round (which measures evasion), then by a detector retrained on all rounds so far (which measures recovery).

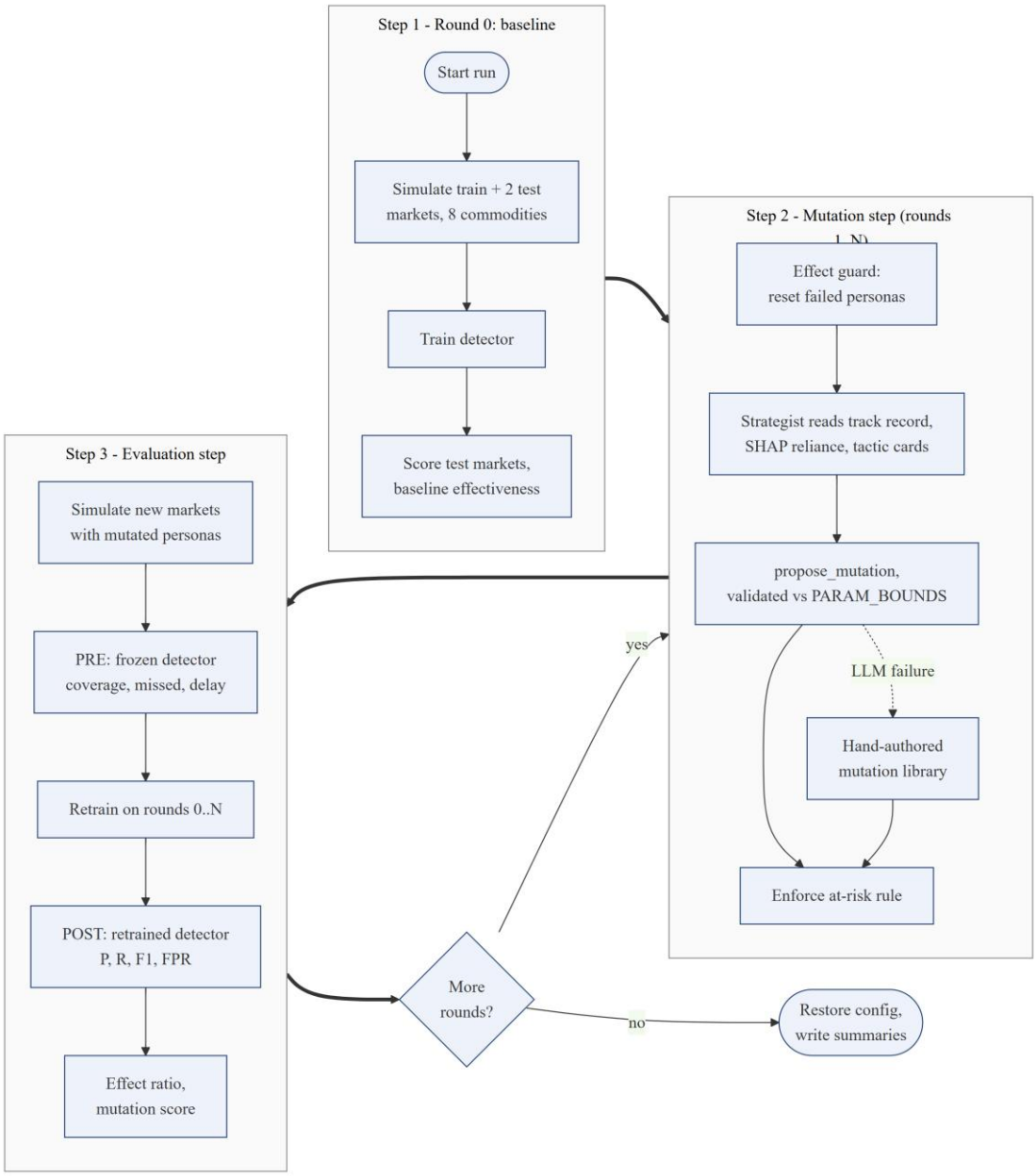



**Figure 4.2: Control flow of the adversarial evaluation loop**

Using two independently simulated test markets per round, rather than one, reduces the chance that a single unusual market drives the round's conclusions. Each round's train and test markets use different seeds and tick offsets, so the detector is never scored on data it was trained on.

## 4.5 CONFIGURATION DESIGN

All behaviour that a user might want to change is in configuration files rather than code. `persona_config.yaml` holds a `round_0` baseline for each persona; mutations are written as top-level persona keys that override the baseline. A single mapping table

(`PERSONA_PARAM_MAP`) translates persona parameter names into simulator settings, so the strategist, the hand-authored library and the simulator all use the same names. `detector_config.yaml` sets the window length and step, the Isolation Forest contamination, the XGBoost hyperparameters and the threshold search range. `llm_config.yaml` selects the LLM provider and sets token limits, the budget cap and call spacing. The pipeline copies the baseline configuration at the start of a run and restores it at the end, even if the run is interrupted, so one experiment's mutations can never leak into the next.

## 4.6 DASHBOARD DESIGN

The dashboard is a read-only consumer of run artefacts (Figure 4.3). A cached data-loader layer reads the files of the selected run, round and test market. Four tabs present the data: Market Overview, Flagged Trades, Model Performance and Analyst Chat. Only two actions call the LLM: generating a case note for a flagged account, and asking the analyst assistant a question. If the LLM is unavailable, those two features show an explanation and everything else continues to work.

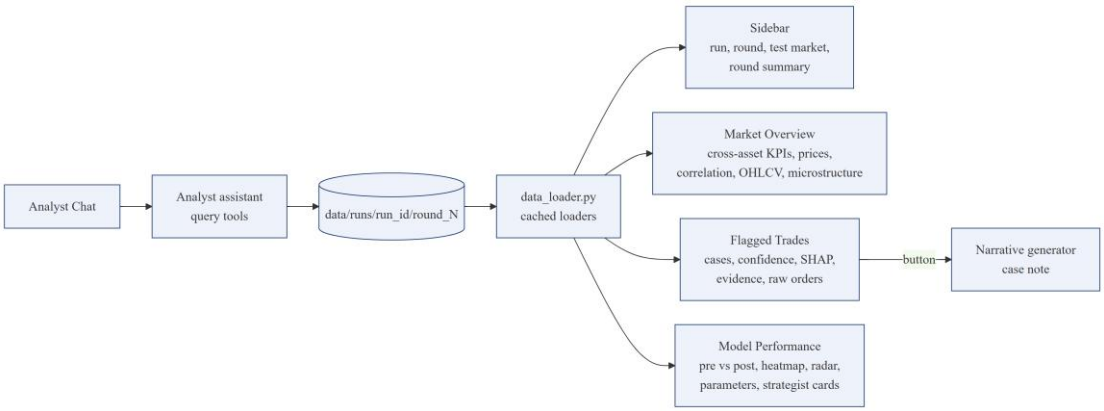



**Figure 4.3: Data flow of the Streamlit dashboard**

## 4.7 KEY DESIGN DECISIONS

**Table 4.3: Design decisions and their rationale**

| Decision | Alternative considered | Reason |
|---|---|---|
| Trader-level detection over 3-minute windows | Market-level (whole-book) detection | Analysts investigate accounts, not markets; account-level labels are exact in simulation |
| One XGBoost classifier per persona | One multi-class model | Persona-specific metrics, thresholds and SHAP explanations (SRS FR-2.3, FR-2.6) |
| OR ensemble of classifiers and Isolation Forest | Weighted score blend | Unknown or mutated schemes can still be caught by the anomaly layer |
| Frozen (pre-retrain) and retrained (post-retrain) scoring | Retrained scoring only | Evasion can only be measured against a detector that has not yet seen the mutation |
| Evasion multiplied by effect ratio | Evasion alone | Prevents the adversary from “evading” by not manipulating |

| Decision | Alternative considered | Reason |
|---|---|---|
| Any-alert coverage in the adversary's score | Persona-specific coverage | Being flagged under another persona's label is still being caught |
| Single validated tool for the LLM | Many read tools in an agent loop | Context is pre-fetched once; fewer calls, lower cost, no unvalidated actions |
| Legitimate look-alike traders | Only simple legitimate agents | Without them every large or heavily cancelling account is trivially anomalous |

# Chapter 5
# Implementation: Agent-Based Market Simulator

This chapter describes Module 1, the simulator that produces every market used in the project. Its purpose is to generate order-book data that behaves like a real futures market, contains manipulation whose timing and accounts are known exactly, and also contains legitimate behaviour that can be mistaken for manipulation. The simulator is a discrete-time agent-based model: one tick represents one second, and in each tick every agent may submit orders to a central limit order book.

## 5.1 LIMIT ORDER BOOK AND MATCHING ENGINE

The order book (`src/core/order_book.py`) keeps resting buy and sell orders in price-level dictionaries, each level holding a queue of orders in arrival order. A new limit order is matched against the opposite side while its price crosses the best opposite quote; each match executes at the resting order's price, and the earliest order at the best price is filled first. This is price-time priority, the matching rule used by futures exchanges. Any unfilled remainder rests in the book. Every trade records the aggressor side (whether the incoming order was a buy or a sell), which several detection features need. Prices are snapped to each commodity's real tick size, so a gold order can only be placed in steps of 0.10 and a natural gas order in steps of 0.001.

Before agents act, the simulator builds a read-only snapshot of the book, `BookState`. It contains the best bid and ask, the mid price, depth at each level, and the visible volume on each side within ±0.2% of the mid. From those volumes the snapshot computes the *order-book imbalance*,

$$\textit{imbalance} = (\textit{bid volume} - \textit{ask volume}) / (\textit{bid volume} + \textit{ask volume})$$

which ranges from −1 (all visible depth on the ask side) to +1 (all on the bid side). Real liquidity providers lean their quotes on this quantity, and that is precisely what spoofing and layering exploit. The snapshot also carries an activity signal (traded volume over the last 60 ticks compared with a slow exponential moving average), which personas use to decide whether the market is busy. Order identifiers are deterministic strings built from a run prefix, the trader id and a counter, so that two runs with the same seed produce byte-identical logs.

## 5.2 CALIBRATION TO REAL MARKET DATA

A simulator is only useful for surveillance research if its prices behave like the real contracts. Veridex takes three kinds of information from real markets: the statistical behaviour of each commodity's price (volatility, drift and mean reversion), estimated from historical futures prices downloaded with the yfinance library; the exchange contract specifications (tick size and price units); and the stylized facts that any realistic market must reproduce, which are checked after simulation (Section 5.7).

### 5.2.1 DATA SOURCE

The script `scripts/fetch_calibration_data.py` downloads five years of daily prices and volumes (14 September 2021 to 14 September 2026, about 1,257 trading days per contract) for the eight futures in Table 5.1. The tickers ending in `=F` are Yahoo Finance's continuous front-month futures series: on each day they show the contract closest to expiry, rolling to the next contract as expiry approaches. These are futures prices, not shares of commodity companies. The script applies three data-quality rules: an empty download stops the process instead of continuing silently; day-to-day moves above 15% are flagged as possible roll artefacts for review instead of being deleted; and missing trading days are not forward-filled, because filling gaps with the last price would understate volatility.

**Table 5.1: Commodity futures used for calibration**

| Commodity | Yahoo ticker | Exchange | Price unit | Tick size (contract spec.) |
|---|---|---|---|---|
| WTI crude oil | CL=F | NYMEX | US$ per barrel | 0.01 |
| Brent crude oil | BZ=F | NYMEX | US$ per barrel | 0.01 |
| Gold | GC=F | COMEX | US$ per troy ounce | 0.10 |
| Silver | SI=F | COMEX | US$ per troy ounce | 0.005 |
| Natural gas | NG=F | NYMEX | US$ per MMBtu | 0.001 |
| Copper | HG=F | COMEX | US$ per pound | 0.0005 |
| Wheat | ZW=F | CBOT | US cents per bushel | 0.25 ¢ (US$ 0.0025) |
| Corn | ZC=F | CBOT | US cents per bushel | 0.25 ¢ (US$ 0.0025) |

### 5.2.2 ESTIMATED PARAMETERS AND THEIR MEANING

The calibration engine (`src/core/calibration.py`) models the logarithm of the price as an Ornstein–Uhlenbeck (mean-reverting) process and estimates its parameters with the standard discrete-time method (Vasicek, 1977): an ordinary least-squares regression of each day's log price on the previous day's,

$$log\ P_{t+1} = \alpha + \beta\ log\ P_t + \varepsilon_t\ ,$$

from which the mean-reversion speed is $\kappa = -\ln \beta$, the long-run level is $\theta = \alpha / (1 - \beta)$, the daily volatility $\sigma$ is the standard deviation of the residuals $\varepsilon_t$, and the drift $\mu$ is the mean daily log return. Volume statistics are taken from the daily volume column. Table 5.2 explains what each parameter means and how the simulator uses it.

**Table 5.2: Parameters extracted from the price data**

| Parameter | How it is estimated | What it means | How the simulator uses it |
|---|---|---|---|
| Daily volatility $\sigma$ | Std. deviation of the AR(1) residuals of log price | The typical size of a day's price move (e.g. 2.2% means a typical daily move of about ±2.2%) | Drives the fundamental value; per tick it becomes $\sigma / \sqrt{23{,}400}$, since a simulated day has 23,400 one-second ticks. It also sets the jump size |
| Drift $\mu$ | Mean daily log return | Slow average trend of the price | Added to the fundamental value each tick as $\mu / 23{,}400$; negligible within one simulated market |
| Mean-reversion speed $\kappa$ | $-\ln \beta$ from the regression | How strongly prices are pulled back to their long-run level; 0 is a random walk | The pull term of the fundamental value. At these values it is almost invisible within a 2.8-hour market |
| Long-run level $\theta$ | $\alpha / (1 - \beta)$ | The level the price reverts towards | Normalised so that the fundamental starts at the commodity's initial price |
| Mean daily volume and its coefficient of variation | Mean and std/mean of daily volume | Typical activity and how much it varies from day to day | Not used directly (see below); activity comes from agent arrival rates |
| Tick size | Exchange contract specification | Smallest allowed price step | Every order price is rounded to it; it also sets the size of the bid-ask bounce |

### 5.2.3 FITTED VALUES AND THE VALUES USED IN THE EXPERIMENTS

Table 5.3 compares the parameters fitted to the five-year yfinance sample with the values the simulator actually used in the experiments reported in Chapter 8. The experiments used long-run constants from published commodity studies (Erb & Harvey, 2006; Geman, 2005; Deaton & Laroque, 1996), which the calibration engine provides as defaults; the yfinance fit is computed and stored by the same engine but is not wired into the simulation runs. The fitted volatilities are higher for every commodity because the 2021–2026 window contains unusually turbulent periods: the 2022 energy shock, which moved crude oil and natural gas sharply, and the strong metals rally of 2024–2026 (gold ended the sample at about US$ 4,350 per ounce and silver at about US$ 64). The long-run constants describe more typical conditions, which suits a study of manipulation in ordinary markets, but switching the simulator to the fitted values is a one-line change and is listed as future work.

**Table 5.3: Parameters fitted from yfinance (Sep 2021 to Sep 2026) and the values used by the simulator**

| Commodity | Daily vol. (fit) | Daily vol. (used) | Drift (fit) | Drift (used) | κ (fit) | κ (used) | Initial price (used) |
|---|---|---|---|---|---|---|---|
| WTI crude oil | 2.61% | 2.20% | 0.030% | 0.005% | 0.012 | 0.035 | 75.00 |
| Brent crude oil | 2.49% | 2.10% | 0.031% | 0.005% | 0.012 | 0.033 | 82.00 |
| Gold | 1.19% | 0.94% | 0.070% | 0.010% | 0.010 | 0.050 | 2,400.00 |
| Silver | 2.55% | 1.50% | 0.078% | 0.010% | 0.001 | 0.060 | 28.50 |
| Natural gas | 5.33% | 4.00% | −0.047% | −0.010% | 0.009 | 0.100 | 2.50 |
| Copper | 1.78% | 1.20% | 0.032% | 0.020% | 0.004 | 0.040 | 4.20 |
| Wheat | 2.20% | 1.60% | 0.005% | 0.002% | 0.006 | 0.080 | 5.80 |
| Corn | 1.62% | 1.40% | 0.004% | 0.002% | 0.003 | 0.080 | 4.30 |

*κ is per trading day. Wheat and corn prices are quoted in US$ (the exchange quotes cents per bushel).*

Three further choices follow from the data. First, the fitted mean-reversion speeds are very small (0.001–0.012 per day, close to a random walk), and even the larger constants (0.03–0.10 per day) have no visible effect within a single 10,000-tick market of about 2.8 hours, so the choice between them does not affect the experiments. Second, the yfinance volume of a continuous futures series is unreliable: it switches between contracts at each roll, which shows up as day-to-day coefficients of variation of 3.1 to 5.6 for gold, silver and copper. Volume is therefore not copied into the simulator; market activity is produced by the arrival rates of the agents (Section 5.3). Third, initial prices are representative long-run levels rather than the latest prices. Because every detector feature is expressed in relative terms (basis points, ratios and multiples of the median order size), the absolute price level does not change the results.

### 5.2.4 FUNDAMENTAL VALUE PROCESS

Each commodity has a hidden fundamental value that follows the Ornstein–Uhlenbeck process with occasional jumps:

$$dX_t = \kappa(\theta - X_t)\, dt + \sigma\, dW_t + J_t$$

where $W_t$ is Brownian motion and $J_t$ a compound-Poisson jump term (on average four jumps per simulated day). The fundamental value is never traded directly. Only the mean-reversion traders observe it, so the movement of market prices towards value is an emergent result of their trading rather than an imposed formula, and the realised price volatility combines the fundamental's volatility with the noise created by order flow and discrete ticks.

## 5.3 LEGITIMATE MARKET PARTICIPANTS

Six types of legitimate agent populate each market (Table 5.4). Their mix follows the broad categories of the CFTC Commitments of Traders report: uninformed retail flow, speculative trend followers and commercial participants with a view on value, together with liquidity providers.

**Table 5.4: Legitimate agent population per market**

| Agent | Count | Behaviour | Role |
|---|---|---|---|
| Noise trader | 20 | With probability 0.6 per tick, a small limit order on a random side, priced around the microprice; unfilled orders expire after a random lifetime | Background liquidity and order flow; cover for manipulation |
| Momentum trader | 8 | Measures the price change over the last 60 seconds; if it exceeds a threshold, sends a marketable order in the trend direction, within a position limit | Speculators; amplifies pumps (exploited by pump-and-dump) |
| Mean-reversion trader | 6 | Trades at the touch against deviations of price from the fundamental value | Commercial participants; anchors price to value |
| Market maker | 1 | Requotes both sides almost every tick; skews quotes against its inventory (in half-spreads) and leans on order-book imbalance | Designated liquidity; the participant that spoofing deceives |
| HFT liquidity provider | 3 | Requotes with probability 0.8 per tick as a ladder of 1–3 levels per side; cancels about 93% of its volume; one-sided when inventory is high | Legitimate heavy canceller and multi-level quoter (hard negative) |
| Execution algorithm | 3 | Works large parent orders (5–20× level depth) by resting big child orders (1–8× depth) at or near the touch, chasing the price, with occasional hedges | Legitimate large-order canceller (hard negative) |

The last two agent types were added specifically to make detection realistic. Without them, the only accounts that ever placed large orders or cancelled most of their orders were manipulators, so even a simple anomaly detector found every spoofer regardless of its tactics. Real futures markets contain many legitimate firms that cancel over 90% of their orders and many institutions that rest and withdraw large orders, which is why regulators have to look for more specific patterns. Section 5.6 shows the effect of these agents on the feature distributions.

Manipulator accounts also behave like ordinary small traders outside their manipulation window (a background order with probability 0.05 per tick, kept close to flat). Without this, a detector could learn “this account only exists while manipulating” rather than the manipulation behaviour itself.

## 5.4 SIMULATION LOOP

Figure 5.1 shows the work done in every tick. The fundamental value is advanced first. Orders whose lifetime has expired, and orders whose cancellation a persona scheduled,

are cancelled next. The market maker then cancels and replaces its quotes. All other actors (noise, momentum and mean-reversion traders, HFT liquidity providers, execution algorithms and the persona controllers) then act in a random order drawn from a seeded generator, so no participant systematically acts first. Each order is matched immediately, trades and order events are logged, inventories and price histories are updated, and the activity signal is refreshed.

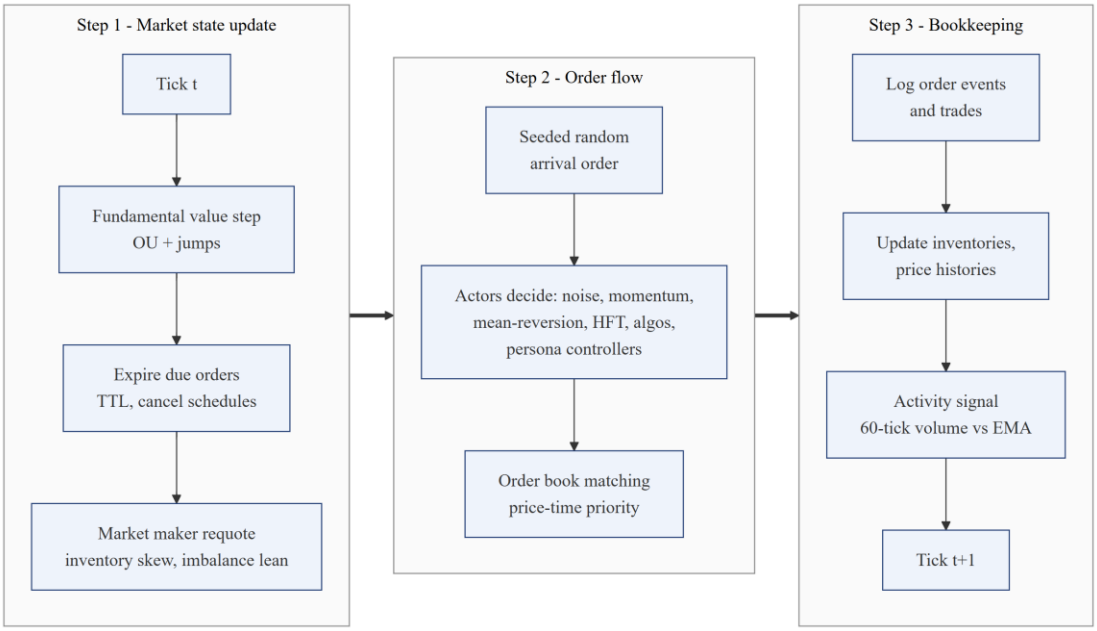



**Figure 5.1: Work performed by the simulator in each tick**

After the last tick, trades are resampled into OHLCV bars, the manipulation episodes are written as ground-truth rows (one row per episode, listing every account involved and the exact parameters used), and all four outputs are validated and written as Parquet files. Eight commodities are simulated in parallel processes, each with its own seed derived from the run seed, so prices in different commodities do not move in lockstep.

## 5.5 MANIPULATION PERSONAS

A persona is a controller that operates one or more manipulator accounts during an episode. Episodes are scheduled at random times in each market; each commodity has an affinity profile that sets how likely each persona is, and each persona has a typical duration (for example 140–420 ticks for spoofing and 220–600 ticks for wash trading). Personas place real orders into the same book as everyone else, so their market effect is genuine. Every persona has a baseline (round 0) parameter set that reflects the documented, unsophisticated version of the scheme, and a set of “knobs” with hard

bounds that the adversarial strategist may change. Orders carry simulator-only audit tags (for example `s` for a spoof leg and `g` for a genuine leg) that the effectiveness accounting uses; the detector never reads them.

### 5.5.1 SPOOFING

Each spoofing cycle has two legs. A small *genuine* order is placed at the best price on one side, for example a sell at the best ask. A large *spoof* order is placed on the other side, slightly behind the best bid, sized at a multiple of the average visible level depth (8× in the baseline). The spoof sits inside the ±0.2% band, so the book suddenly looks bid-heavy: the market maker leans its quotes upward and noise traders price off a higher microprice. The price drifts towards the genuine sell, which fills, and a few ticks later the spoof is cancelled before anyone can trade against it. Table 7.1 (Section 7.3) lists its knobs. They encode evasion tactics from the enforcement cases: randomised size and cancel timing (Coscia), splitting the spoof across levels, using a second account for the spoof leg (JPMorgan), keeping the spoof pinned behind a moving price (Sarao), trading aggressively after pulling the spoof (Oystacher), acting mainly in busy periods, and trading genuinely alongside the spoof (cover trading).

### 5.5.2 LAYERING

Layering, modelled on the Tower Research case, places a small genuine order at the best price on one side and several moderate orders on consecutive price levels on the other side, all cancelled together after the genuine order fills. No single layer is extreme; the deception is the *shape* of the depth. Layer $i$ sits $(i+1) \times$ spacing behind the best price, so the spacing directly controls whether the layers count as near-touch depth. Measurements made during development showed that a spacing of 0.0004 keeps the full effect, 0.0005 keeps 88%, and 0.001 keeps only about 42% because the layers fall outside the band that other participants watch. Its knobs are in Table 7.2.

### 5.5.3 WASH TRADING

Wash trading uses groups of colluding accounts. In each cycle one account rests an order strictly inside the spread and a partner immediately takes it at the same price and size, so no outsider can trade in between (if the spread is only one tick wide, the group waits). Roles rotate around a ring (A sells to B, B to C, C to A), so every account's net

position stays close to zero while gross volume prints to the market. Several groups can split the persona's total cross rate, which dilutes per-account statistics without adding volume. Camouflage cycles replace some crosses with genuine trades with the market (knobs in Table 7.3).

### 5.5.4 PUMP-AND-DUMP (MOMENTUM IGNITION)

The pump-and-dump persona is a coalition of accounts that moves through four phases (Figure 5.2). During *accumulation* the accounts quietly buy with passive bids. In the last 30% of the burst they *ignite*: they send marketable buys through the ask, which moves the price and triggers the momentum traders. During *distribution* they offer their inventory at the ask into the demand they created, and finally they *dump* whatever remains with aggressive sells. Staggering gives each account its own random start, decoys are small trades in the opposite direction, and the busy-period bias delays action until the market is active (knobs in Table 7.4).




**Figure 5.2: Phases of the pump-and-dump persona**

### 5.5.5 COVER TRADING

Real spoofers such as Sarao and Coscia also traded genuinely while they manipulated, so their manipulative orders were a minority of their activity. With `cover_rate` above zero, every spoofing or layering account keeps sending genuine orders during the

episode: 35% are marketable and the rest rest at the account's own side of the touch, and the account leans against its position so it stays roughly flat. `cover_size` sets their size relative to the level depth. Because the detector's cancel, fill and large-order ratios are weighted by volume, only cover volume comparable to the scheme's own volume dilutes them; the depth reference is fixed when cover trading starts, so the account's own resting cover orders cannot inflate the size of later ones. Cover trading costs the manipulator inventory risk, and its genuine resting orders partly offset the fake depth.

## 5.6 BEHAVIOURAL PROFILES OF LEGITIMATE AND MANIPULATIVE ACCOUNTS

Table 5.5 compares the median feature values of each account type in a crude-oil market during development. The high-frequency liquidity providers cancel 93% of their volume, quote on several levels and often have orders cancelled on one side while they trade on the other, so cancellation and layering features alone no longer separate manipulators from legitimate firms. The execution algorithms place orders up to 36 times the median order size and cancel about two thirds of their volume. Spoofers remain more extreme in size, which leaves the adversary a real but costly option: smaller spoofs look more like algorithm orders but create less fake depth.

**Table 5.5: Median account-window feature values by account type (crude oil, development measurement)**

| Account type | Avg order size | Max order size | Large-order share | Cancel ratio | Fill ratio | Layered share | Spoof signature |
|---|---|---|---|---|---|---|---|
| Other legitimate | 0.6× | 1.9× | 0.00 | 0.21 | 0.79 | 0.00 | 0.00 |
| HFT provider | 3.3× | 5.7× | 0.39 | 0.93 | 0.07 | 0.43 | 0.47 |
| Execution algo | 13.1× | 35.9× | 0.99 | 0.68 | 0.31 | 0.00 | 0.20 |
| Layering | 10.3× | 24.0× | 0.98 | 0.98 | 0.02 | 0.98 | 0.92 |
| Spoofing | 34.7× | 103.7× | 0.99 | 0.99 | 0.01 | 0.00 | 0.96 |

*Sizes are relative to the commodity's median order size; ratios are volume-weighted.*

## 5.7 REALISM CHECKS

Table 5.6 reports stylized facts of one-minute (60-tick) log returns in the round-0 test market of the experiment, together with the simulated one-minute volatility and the target implied by the calibrated daily volatility (daily volatility divided by $\sqrt{390}$). Most commodities show heavy tails (positive excess kurtosis) and volatility clustering (positive autocorrelation of absolute returns), the two most robust stylized facts of real markets. The order-to-trade ratio of about 2.0–2.4 and the overall cancel ratio of about 50% are within the range seen in electronic futures markets.

**Table 5.6: Stylized facts of 60-tick returns, round 0 test market**

| Commodity | Excess kurtosis | ACF(1) of \|r\| | ACF(1) of r | Sim. vol ($\times 10^{-3}$) | Target vol ($\times 10^{-3}$) | Order-to-trade | Cancel ratio |
|---|---|---|---|---|---|---|---|
| Brent crude | 1.74 | 0.166 | 0.112 | 1.09 | 1.06 | 2.22 | 0.54 |
| Copper | 10.68 | 0.234 | −0.084 | 1.05 | 0.61 | 2.36 | 0.57 |
| Corn | −0.21 | 0.036 | −0.416 | 1.50 | 0.71 | 2.08 | 0.51 |
| WTI crude | 3.98 | 0.261 | −0.096 | 1.38 | 1.11 | 2.19 | 0.53 |
| Gold | 8.21 | 0.210 | 0.231 | 0.50 | 0.48 | 2.37 | 0.57 |
| Natural gas | −0.23 | 0.058 | 0.106 | 2.36 | 2.03 | 2.03 | 0.49 |
| Silver | 6.34 | 0.261 | 0.016 | 1.25 | 0.76 | 2.20 | 0.53 |
| Wheat | 5.72 | 0.052 | −0.241 | 1.55 | 0.81 | 2.12 | 0.52 |

The checks also reveal limitations that are reported rather than hidden. Simulated volatility is within about 25% of the target for Brent, WTI, gold and natural gas, but between 1.6 and 2.1 times too high for copper, silver, wheat and corn, because microstructure noise from discrete ticks and small order sizes adds to the fundamental volatility in markets where the tick is large relative to the daily move. Corn and wheat show strongly negative first-order return autocorrelation (bid-ask bounce), and corn and natural gas show no excess kurtosis in this sample. These differences do not affect the detection experiments, which compare behaviour within the same markets, but they would need further tuning for studies of price dynamics.

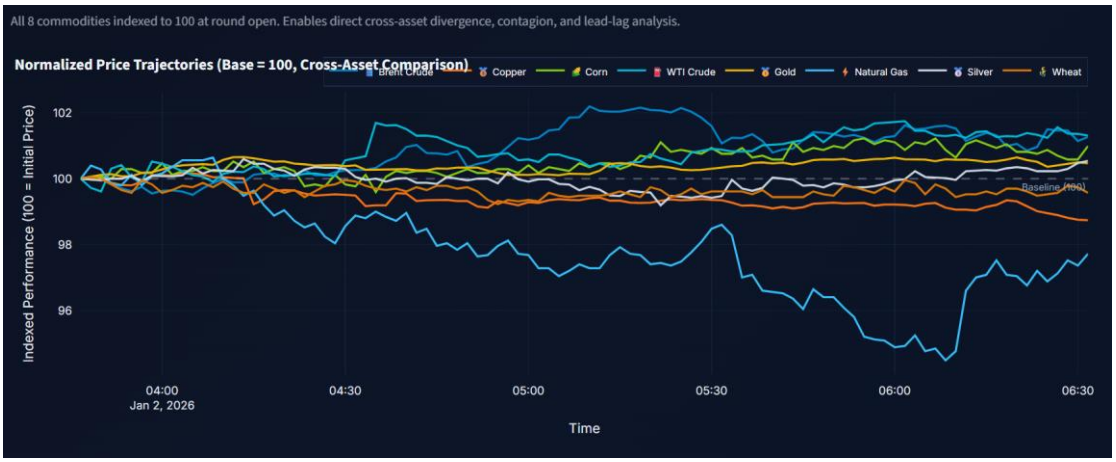



**Figure 5.3: Normalised price paths of the eight commodities (dashboard, round 5 test market)**

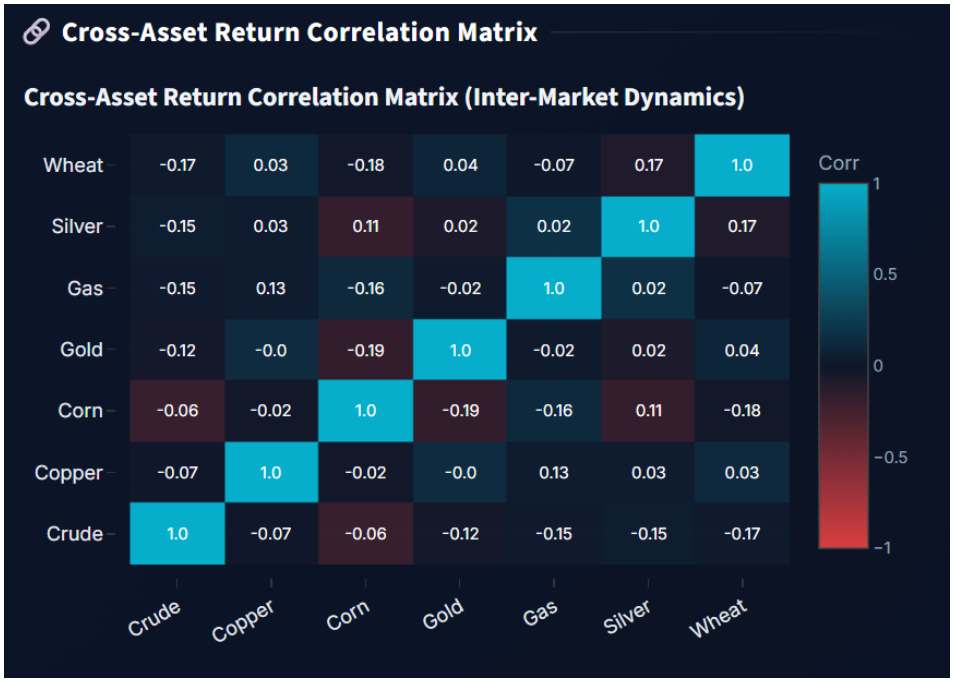



**Figure 5.4: Cross-asset return correlation matrix (dashboard, round 5 test market)**

Figure 5.3 shows the eight simulated price paths indexed to 100. Each commodity moves independently, with the expected differences in volatility (natural gas is the most volatile). Figure 5.4 confirms that returns are nearly uncorrelated across commodities (absolute correlations below 0.2), as intended: each market is simulated with its own seed, and cross-market linkages are outside the scope of this project.

# Chapter 6
# Implementation: Detection Engine and Adversarial Loop

This chapter describes Module 2, which turns raw order-book logs into alerts with explanations, and Module 3, which runs the adversarial experiment and measures both evasion and effectiveness. Figure 6.1 gives an overview of the detection pipeline.

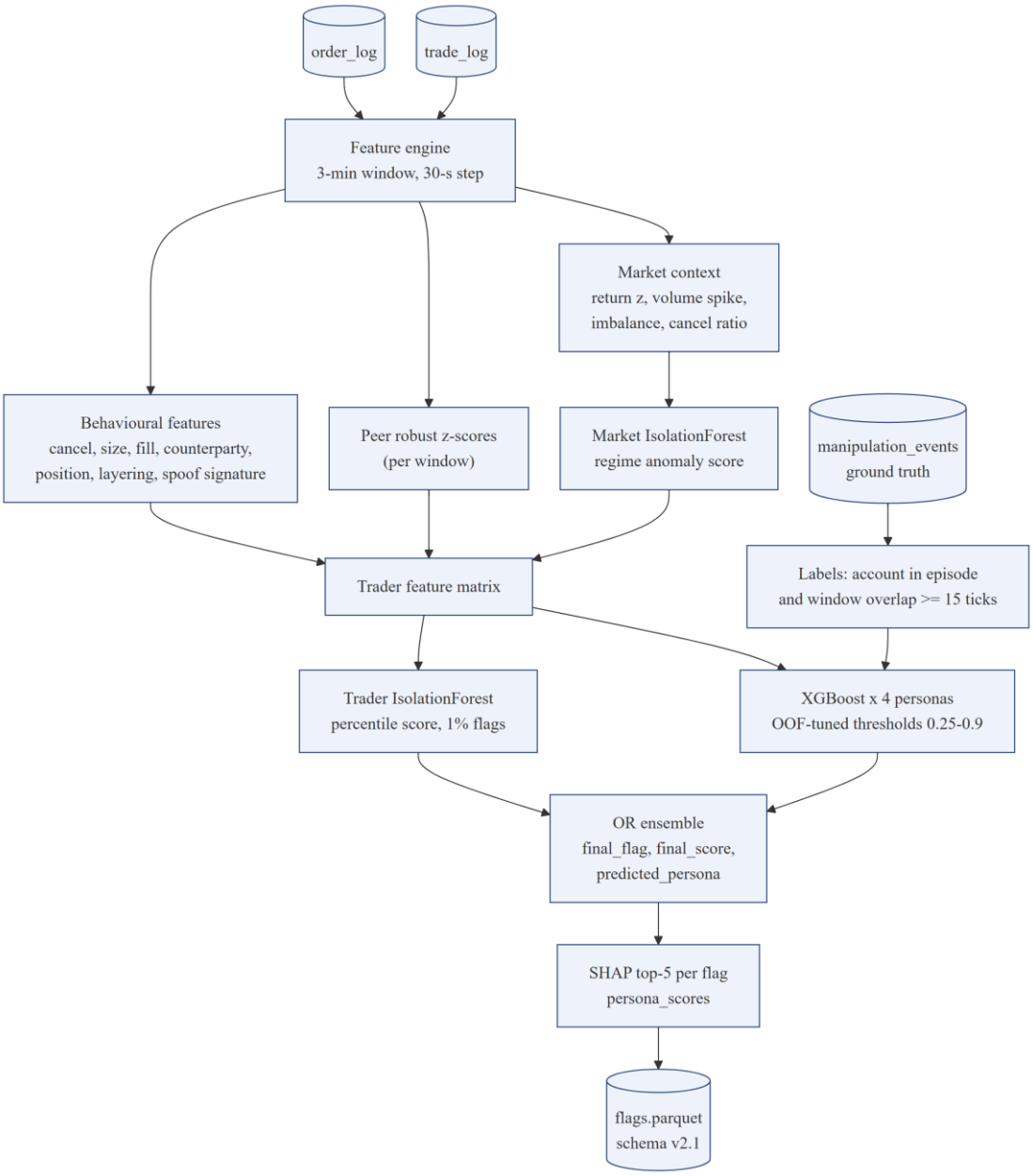



**Figure 6.1: Detection pipeline from logs to explained flags**

## 6.1 UNIT OF DETECTION

The detector scores *account-windows*: one row for each trader account that was active in a three-minute window of one commodity's market. Windows advance in 30-second

steps, so consecutive windows overlap and an episode of a few minutes is seen by several windows. Trader-level rows match how surveillance analysts work: they investigate accounts over periods of time, not abstract market states. The designated market maker is excluded from the feature table because exchanges surveil it under separate quoting obligations; the HFT liquidity providers and execution algorithms are *not* excluded, so the detector must learn to tell them apart from manipulators.

## 6.2 FEATURE ENGINEERING

For each account-window the feature engine (`src/detector/features.py`) computes 50 features from the order and trade logs, grouped in Table 6.1, and the market-regime anomaly score described in Section 6.4 is appended as a context feature. Ratios are weighted by volume, so a few tiny orders cannot hide a large one. Sizes are expressed relative to the commodity's median order size, so the same feature works for gold and for natural gas.

**Table 6.1: Feature catalogue**

| Group | Features | What they capture |
|---|---|---|
| Activity | n_orders, n_trades, exec_vol, gross_volume_share, activity_vs_history | How much the account does, and how this compares with its own history |
| Order size | avg_order_size_rel, max_order_size_rel, order_size_cv, large_order_share (orders ≥ 4× median) | Unusually large or irregular orders |
| Cancellation | cancel_ratio, fast_cancel_ratio (≤ 10 s), large_cancel_ratio, mean_cancel_lifetime, n_large_cancels, cancel_side_imbalance, cancel_replace_ratio | Orders placed and quickly withdrawn; cancel-and-replace behaviour |
| Layering | layered_order_share (same tick, same side, ≥ 3 price levels) | The Tower layering fingerprint |
| Execution | fill_ratio, exec_side_imbalance, aggressor_ratio, passive_distance_bp, cancel_exec_opposition | Which side the account actually trades, and how aggressively |
| Spoofing signature | spoof_signature: share of the account's large cancelled orders during which it was filled on the opposite side | The bait-and-switch itself |
| Counterparty and position | top_cpty_share, reciprocal_ratio, n_counterparties, net_flatness, trade_size_cv, price_dev_bp | Trading repeatedly with the same partner while staying flat (wash trading) |
| Position build-up | prev_net_ratio, long_net_ratio, reversal_score, coordinated_peers, signed_flow_return | Building a position, pushing the price, then reversing; other accounts doing the same (pump-and-dump) |
| Market context | mkt_return_bp, mkt_volatility_bp, mkt_range_bp, mkt_volume_spike, mkt_trade_imbalance, mkt_order_imbalance, mkt_cancel_ratio, mkt_abnormal_return_z, mkt_anomaly_score | What the whole market was doing in the window |
| Peer comparison | peer_z_* robust z-scores of eight key features against all accounts in the same window | How unusual the account is relative to its peers right now |

Two features deserve explanation. `spoof_signature` looks at every large order the account cancelled and asks whether, while that order was resting, the account was filled on the opposite side. For an honest trader this almost never happens; for a spoofer it is

the whole strategy. `coordinated_peers` counts other accounts in the same window that traded aggressively in the same direction, which captures the coordination of a pump-and-dump coalition. The peer z-scores use the median and the median absolute deviation rather than the mean and standard deviation, so a single extreme account cannot distort the scale against which everyone else is compared.

## 6.3 GROUND-TRUTH LABELS

Because the simulator logs every episode, labels are exact. An account-window is labelled positive for a persona if the account is one of the episode's accounts and the window overlaps the episode by at least 15 ticks. The overlap condition avoids labelling a window positive because of one or two seconds of overlap at its edge. Each persona has its own label column, so the same account-window can be evaluated separately by each classifier.

## 6.4 UNSUPERVISED LAYER: ISOLATION FORESTS

Two Isolation Forests, implemented with scikit-learn (Pedregosa et al., 2011), are trained without labels.The *market-regime forest* is trained on the market-context features of each window (contamination 5%); its anomaly percentile becomes the `mkt_anomaly_score` context feature. The *trader forest* is trained on the behavioural features of every account-window (200 trees, contamination 1%). Its inputs are robustly scaled, and its raw scores are converted into percentiles of the training-score distribution, so a score of 0.99 means “more anomalous than 99% of training rows” and the flag threshold is simply 0.99. The trader forest is persona-agnostic: it flags accounts that are unusual for any reason, which lets it catch schemes the supervised layer has not seen, at the cost of also flagging unusual legitimate accounts.

## 6.5 SUPERVISED LAYER: PER-PERSONA XGBOOST CLASSIFIERS

For each persona a separate XGBoost binary classifier predicts whether an account-window belongs to that persona. Separate models give persona-specific metrics, thresholds and explanations, which a single multi-class model would blur. Positive rows are rare, so each model weights positives by the ratio of negatives to positives

(capped at 50). Table 6.2 lists the hyperparameters; a smaller variant is used when very few positive rows are available.

**Table 6.2: XGBoost hyperparameters**

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| n_estimators | 300 (150 small) | subsample | 0.8 |
| max_depth | 4 (3 small) | colsample_bytree | 0.8 |
| learning_rate | 0.05 | gamma | 0.5 |
| min_child_weight | 1 | reg_alpha / reg_lambda | 0.1 / 2.0 |
| eval_metric | logloss | scale_pos_weight | n_neg / n_pos, capped at 50 |

**Threshold tuning.** A default threshold of 0.5 is rarely right for rare classes. Each classifier's decision threshold is chosen from a grid of 27 values between 0.25 and 0.9 to maximise the F1 score on *out-of-fold* predictions. The folds are formed with GroupKFold over 600-tick time blocks, so overlapping neighbouring windows always fall in the same fold; otherwise information would leak between training and validation through the overlap. Keeping the threshold inside [0.25, 0.9] protects against extreme values when a round contains only a few episodes.

## 6.6 ENSEMBLE, PERSONA ASSIGNMENT AND SHAP EXPLANATIONS

An account-window is flagged if any persona classifier exceeds its threshold *or* the trader Isolation Forest fires (an OR ensemble). The flag records which layers triggered, a final score (the highest persona probability, or the anomaly percentile if that is higher), and a predicted persona: the persona whose probability exceeds its own threshold by the largest margin, or “anomaly” if only the Isolation Forest fired. Every flagged row also stores all four persona probabilities (so the dashboard can show the confidence for each persona) and the five largest SHAP contributions of the classifier that produced the flag, computed exactly with TreeSHAP. The result is written to `flags.parquet`, schema version 2.1.

## 6.7 EVALUATION METRICS

Metrics are always reported per persona, never as one combined number. At row level the detector is evaluated with precision, recall, F1 and false-positive rate, using each classifier's tuned threshold. At episode level three further measures are computed:

- **Events detected and detection latency**: an episode is detected if any of its accounts is flagged in a window overlapping the episode; latency is the number of ticks from the episode's start to the end of the first flagged window.
- **Episode coverage**: for each episode, the share of its labelled account-windows that were flagged, averaged over episodes. Unlike row recall, coverage does not change when a mutation simply adds more accounts to an episode.
- **Any-alert variants** (caught coverage, caught events, caught latency), computed on the final flag rather than the persona's own classifier. An account flagged under another persona's label, or only by the anomaly layer, is still caught. These are the measures used to score the adversary.

## 6.8 ADVERSARIAL ROUND ORCHESTRATION

The round orchestrator (`src/adversarial/orchestrator.py`) runs each round as follows. It simulates one training market and two test markets, each covering all eight commodities, with different seeds and tick offsets. It loads or computes their features and labels (features are cached next to the logs, so cumulative retraining does not recompute earlier rounds). It then scores the test markets with the detector as it stood at the end of the previous round, the *pre-retrain* or *frozen* evaluation. It retrains the detector on the training markets of all rounds so far, scores the test markets again (*post-retrain*), builds and writes the flags, measures the effectiveness of every manipulation episode, and saves everything to the round's `metrics.json`. The frozen evaluation answers “did the new tactic evade the detector that existed when it was introduced?”; the retrained evaluation answers “can the detector learn to catch it?”.

## 6.9 MANIPULATION EFFECTIVENESS AND THE ADVERSARY'S OBJECTIVE

An evasion only matters if the manipulation still works. The effectiveness module measures, for every episode and from the test market's own logs, one primary quantity per persona (Table 6.3). The median across episodes is compared with the round-0 baseline to give the *effectiveness ratio*.

**Table 6.3: Primary effectiveness measures**

| Persona | Primary measure | Definition |
|---|---|---|
| Spoofing | fake_depth_share | Share of the visible near-touch depth (±0.2% band) on the spoofed side that the spoof orders made up when placed |

| Persona | Primary measure | Definition |
|---|---|---|
| Layering | fake_depth_share | The same measure for the stacked layers |
| Wash trading | wash_volume_share | Self-crossed volume divided by all market volume during the episode |
| Pump-and-dump | ignition_pressure | Peak share of market volume taken by the coalition's marketable buying in any 60-tick window |

These measures were chosen after simpler ones proved too noisy. The realised price move of a spoof, for example, varied between −0.25× and 5.3× of baseline from one round to the next because it is dominated by unrelated price noise, whereas the fake-depth share measures the lever the spoofer actually pulls. The adversary's objective for each persona combines evasion of the frozen detector with the effectiveness ratio:

$$evasion = 0.5 \times (1 - coverage) + 0.3 \times missed + 0.2 \times \min(1, delay / 300)$$

$$score = evasion \times \min(1, effect\ ratio), \quad and\ score = 0\ if\ effect\ ratio < 0.5$$

Here *coverage* is the frozen detector's any-alert episode coverage, *missed* is the share of episodes never flagged at all, and *delay* is the mean detection latency of caught episodes in ticks (capped at 300, about five minutes). An effect above the baseline earns nothing extra, because a bigger scheme is easier to catch and would otherwise be rewarded for the wrong reason. An effect below half of the baseline scores zero: that is not evading, it is not manipulating.

## 6.10 THE EFFECT GUARD

During development the LLM strategist sometimes made changes that quietly destroyed a scheme's effect (for example, widening the layer spacing until the layers no longer counted as depth) and then failed to undo them for several rounds. The effect guard enforces the effectiveness constraint in code rather than relying on the LLM. Before each mutation step it classifies every persona by its last effectiveness ratio:

- **Failed** (ratio below 0.5): the persona's parameters are reset to its best effective earlier set, and the strategist is told which changes broke it and not to re-apply them.
- **At risk** (ratio below 0.65): the strategist is warned and shown the effect-costing changes since the last comfortably effective round. After its proposal, any *new* effect-costing change is undone, and if it undid none of the earlier ones, they are all reverted. Evasion-only changes such as cancel timing are kept.

- **Surplus** (ratio at least 1.5): the strategist is told that the extra effect earns nothing and can be spent on evasion, for example with knobs it has never tried.

The direction in which each knob lowers effectiveness is recorded in a small table (for example, wider layer spacing, fewer layers and smaller layers all reduce fake depth). In the six-round experiment reported in Chapter 8, spoofing entered the at-risk band after round 3 (0.62×). Following the warning, the strategist raised the spoof size from 4× to 5× level depth in round 4 (its stated reason was “undo prior down move, restores effect”), so the guard had nothing to revert and the effect recovered to 0.74×. No persona fell below the 0.5 floor, so no reset was needed. The reset and revert paths are covered by unit tests built from scenarios observed in earlier development runs.

# Chapter 7
# Implementation: LLM Layer and Dashboard

This chapter describes Module 4, the LLM and agentic layer, and Module 5, the Streamlit dashboard. The LLM is used in three bounded roles: as the adversarial strategist, as a case-note writer, and as an analyst assistant.

## 7.1 LLM ACCESS

All three roles call the model through one client (`src/agentic/llm_client.py`, built on LiteLLM). A single setting selects the provider for the whole project; the experiments used the open-weight gpt-oss-120b model hosted by Cerebras, with Google Gemini as an alternative. Because the project had a small, fixed amount of API credit, the client enforces a hard budget cap, answers repeated requests from a cache, limits the number and rate of calls and stops retrying after unrecoverable errors, and every call is logged for auditing. If the LLM is unavailable, the strategist falls back to a hand-written mutation library and the dashboard hides only the two LLM features.

## 7.2 ADVERSARIAL STRATEGIST

Before every round after round 0, the strategist (`src/agentic/adversarial_strategist.py`) proposes new parameters for each persona. The design goal was a bounded agent that produces useful mutations from as few calls as possible. Instead of letting the model explore the run with many read-only tool calls, which would multiply the cost, the system pre-fetches a compact context once per round and offers the model a single tool, `propose_mutation`. For each persona the context contains:

- every knob with its type, bounds, current value and a one-line meaning;
- the full track record: for every earlier round, the parameters used, who chose them, the frozen detector's any-alert coverage, episodes caught and delay, the retrained recall, the effect ratio and the score;
- knobs never changed so far and knobs currently at a bound, as exploration hints;
- the detector's current reliance (mean absolute SHAP of its six most important features) and the features whose importance rose since the last round;

- the tactic cards for the persona, and an effect-guard note (failed, at risk or surplus) when applicable.

The system prompt states the objective formula from Section 6.9, explains that being flagged under another persona's label still counts as caught and that effect above 1.0× earns nothing, and prescribes a method: read the track record, start from the best parameter set, change one to three knobs aimed at features the detector currently relies on, prefer low-cost tactics, try untried knobs when the score stalls, and never resubmit a past parameter set. Each proposal is validated against the hard bounds; a rejected proposal is returned once with the exact validation error, and a second rejection (or any LLM failure) falls back to the hand-authored mutation library for that round. The loop is limited to five model turns per round.

### 7.2.1 TACTIC CARDS

Tactic cards (`docs/tactic_cards.yaml`) are short, structured descriptions of documented evasion tactics: the source case, the knobs that implement it, the detector features it is expected to weaken, and what it costs the manipulator. For example, the spoofing card “Cancel-timing randomisation” cites the Coscia case, maps to `cancel_jitter` and `cancellation_delay_ticks`, weakens `fast_cancel_ratio` and `mean_cancel_lifetime`, and warns that longer-resting spoofs risk being filled. Some cards record measured trade-offs from development experiments, such as the hard spacing limit for layering. Keeping the cards short matters because they are sent to a paid model every round.

## 7.3 THE MUTATION SPACE: WHAT THE STRATEGIST CAN CHANGE

A *mutation* is a change to a persona's parameters, its “knobs”, between two rounds. The strategist never writes code or invents new behaviour: it can only move the documented knobs of each persona within hard bounds, and every knob corresponds to a tactic that real manipulators were documented using. This section explains what each knob means, why it matters to the detector, and what happens when it is raised or lowered.

### 7.3.1 WHY MUTATIONS MATTER: EVASION AGAINST EFFECT

Every knob changes the behavioural footprint of the scheme, that is, the values the detector's features take for the manipulating accounts (Section 6.2). Most knobs also change how well the scheme works. The two effects usually pull in opposite directions, and the strategist's objective (Section 6.9) rewards only mutations that reduce detection while keeping at least half of the round-0 market effect. Knobs fall into three groups:

- **Effect-costing knobs** weaken detection but also weaken the manipulation, for example smaller spoofs, wider layer spacing or a lower wash-trading rate. Pushed too far, the scheme stops working and scores zero. These are the knobs the effect guard watches (Section 6.10).
- **Free evasion knobs** change the footprint with little effect on the market effect, for example randomised cancel timing, irregular sizes, spreading activity across more accounts or groups, and waiting for busy periods. They are usually the strategist's first choice; their cost is operational (more accounts to control, fewer cycles when the market is quiet).
- **Effect-raising knobs** strengthen the manipulation but make it more visible, for example larger spoofs or more aggressive ignition. Because effect above 1.0× earns nothing, the strategist only raises these to recover lost effect.

Tables 7.1 to 7.4 describe every knob. In the “if increased” and “if decreased” columns, *harder to detect* and *easier to detect* refer to the features named, and *effect* refers to the persona's primary effectiveness measure (Table 6.3).

**Table 7.1: Spoofing knobs: meaning and effect of changes**

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| order_size_multiplier | Size of the fake order as a multiple of the average visible level depth | 8.0 [1–15] | More fake depth and a stronger push on the book (effect up); extreme max_order_size_rel and large_order_share make it easier to detect | Looks more like a legitimate algorithm's order, so harder to single out; less fake depth. Below about 4× the effect falls steeply (measured 0.64×) |
| cancellation_delay_ticks | How long the fake order rests before it is cancelled | 5 [1–30] | Fewer fast cancels (fast_cancel_ratio down, longer lifetimes), so harder to detect; the resting spoof is more likely to be hit and filled, a real cost | Very fast cancels, the classic spoofing signature (Coscia), easy to detect; the spoof is almost never filled |
| cancel_jitter | Randomness of the cancel delay | 0.0 [0–0.8] | Irregular timing hides the machine-like rhythm that classifiers learn | Regular, scripted timing that is easy to learn |

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| size_jitter | Randomness of the spoof size | 0.0 [0–0.8] | Sizes vary from cycle to cycle, breaking the order-size pattern; average effect roughly unchanged | Identical sizes every cycle, a scripted fingerprint |
| n_split_orders | Number of pieces the spoof is split into, on consecutive price levels | 1 [1–4] | Each piece is smaller (max_order_size_rel down), but several levels at once resemble layering and are often caught under that label; more cancel events | One large order, the single most obvious feature |
| n_accounts | Accounts used: with 2 or more, the genuine and fake legs come from different accounts | 1 [1–3] | Breaks the per-account bait-and-switch (spoof_signature down); the strongest spoofing evasion in the experiment (coverage 0.59) | Both legs in one account, so spoof_signature exposes the scheme directly |
| frequency | Chance per tick of starting a new spoofing cycle | 0.4 [0.05–1.0] | More cycles and more total impact, but more suspicious activity in every window | Less evidence per window and less total impact; the fake depth per cycle is unchanged |
| price_aggressiveness | Distance of the spoof behind the best price (fraction of price) | 0.0005 [0.0001–0.01] | Less likely to be filled, but beyond about 0.2% the order no longer counts as near-touch depth and the effect collapses | Closest to the touch: maximum effect but a higher risk of being filled |
| reprice_to_touch | Keep the spoof pinned behind a moving best price (Sarao) | 0 [0 or 1] | The spoof stays effective as the price moves, but every re-pricing is a cancel-and-replace (cancel_replace_ratio up) | The spoof stays at its original price and drifts away from the touch |
| flip_after_cancel | Chance of an aggressive trade in the genuine direction after the cancel (Oystacher) | 0.0 [0–1.0] | More real executions (fill_ratio up, cancel_ratio down) and profit-taking, at the cost of real trading costs and visible aggressive flow | No flipping |
| busy_period_bias | Probability of skipping ticks when the market is quieter than usual | 0.0 [0–1.0] | Activity hides among naturally high volume (activity_vs_history and market-relative signals weaker); fewer cycles in quiet periods | Acts regardless of conditions and stands out in quiet periods |
| cover_rate | Per-account chance per tick of a genuine cover order | 0.0 [0–0.5] | Spoofs become a minority of the account's flow, diluting size, cancel and fill ratios; costs inventory risk | The account trades only the scheme, so its ratios are extreme |
| cover_size | Cover order size as a multiple of level depth | 0.0 [0–3.0] | Dilutes the volume-weighted ratios more (about 1× depth is needed), but very large cover makes the account an outlier again and offsets the fake depth | Small cover barely changes volume-weighted features |

**Table 7.2: Layering knobs: meaning and effect of changes**

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| n_layers | Orders stacked on consecutive price levels per cycle | 4 [2–8] | A deeper, more convincing fake book (effect up) and a stronger layered_order_share fingerprint | Less fake depth; below 3 levels it no longer looks like layering and is caught as spoofing instead |

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| layer_spacing | Gap between layers as a fraction of price | 0.0002 [0.00005–0.003] | Layers sit further from the touch and are rarely filled, but they leave the ±0.2% band: 0.0005 keeps 0.88× of the effect, 0.001 only 0.42× | All layers near the touch: maximum effect, more likely to be filled |
| layer_size_multiplier | Size of each layer as a multiple of level depth | 2.0 [0.5–6.0] | More fake depth (effect up), but large-order features become extreme | Harder to spot, but thin layers barely change the visible depth (strong effect cost) |
| size_taper | How much smaller deeper layers are (Tower Research) | 0.0 [0–0.9] | A natural-looking profile that weakens max-size features, but deeper layers add less depth | Equal-sized layers, a uniform and easily learned pattern |
| cancellation_delay_ticks / cancel_jitter | How long layers rest and how random the timing is | 5 / 0.0 | Fewer fast cancels and irregular timing, so harder to detect; layers risk being filled | Fast, regular cancels that are easy to detect |
| frequency | Chance per tick of a new layering cycle | 0.3 [0.05–1.0] | More cycles and more impact, more visible activity | Less activity per window and less total impact |
| busy_period_bias | Act mainly when the market is busy | 0.0 [0–1.0] | Hides among high volume; fewer cycles in quiet periods | Stands out in quiet periods |
| cover_rate / cover_size | Genuine cover trading during the episode | 0.0 / 0.0 | Dilutes size, cancel and layering ratios (measured: coverage 0.65 with cover 0.3 × 1.0); inventory risk | Account activity is entirely the scheme |

**Table 7.3: Wash-trading knobs: meaning and effect of changes**

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| trade_frequency | Total rate of self-crosses per tick, split across groups | 0.5 [0.05–1.0] | More artificial volume (effect up), but a conspicuous share of market volume | Less artificial volume, which is the whole purpose of the scheme: the effect falls in proportion (strong cost) |
| n_groups | Independent colluding groups sharing the same total rate | 1 [1–5] | Each account is less active and per-account signals are diluted, at no cost in total wash volume | One group carries all the volume and stands out |
| ring_size | Accounts per group; trades rotate A→B→C→A | 2 [2–6] | Counterparties are spread out (top_cpty_share and reciprocal_ratio down); more accounts to control | 2 is a fixed pair trading only with each other, the textbook pattern |
| size_jitter | Randomness of trade sizes | 0.0 [0–0.9] | Varied, natural-looking trade sizes | Identical scripted lots (low trade_size_cv) |
| camouflage_rate | Share of cycles replaced by a genuine trade with the market | 0.0 [0–0.5] | Breaks the flat-position and counterparty patterns, but adds real market exposure and reduces wash volume | Pure self-trading with a perfectly flat position |
| busy_period_bias | Act mainly when the market is busy | 0.0 [0–1.0] | Artificial volume blends into naturally high volume; fewer crosses in quiet periods | Volume spikes appear in quiet periods |

**Table 7.4: Pump-and-dump knobs: meaning and effect of changes**

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| n_coordinated_accounts | Number of colluding accounts | 3 [2–15] | Each account is smaller and less extreme, with more combined buying power, but more coordinated activity (coordinated_peers) | Fewer, more extreme accounts |

| Knob | What it controls | Baseline [bounds] | If increased | If decreased |
|---|---|---|---|---|
| accumulation_size | Target position per account | 150 [50–1000] | A bigger pump with more profit potential and a larger footprint | Harder to see, but less buying power for the ignition |
| burst_duration_ticks | Duration of accumulation plus ignition | 200 [50–1000] | Activity is spread out and activity spikes are weaker, but the price push is weaker | A short, concentrated and conspicuous burst |
| dump_delay_ticks | Length of the passive distribution phase before the dump | 100 [10–500] | A slower exit with a less abrupt reversal (reversal_score down), but the price may fall back before the coalition has sold | A sharp buy-then-sell reversal, the classic signature |
| stagger | Random start delay per account (fraction of the burst) | 0.0 [0–1.0] | Accounts look like independent traders (coordinated_peers down), but the pump is less synchronised and lifts the price less | All accounts act together, a clear coordination signal |
| ignition_aggression | How far through the ask the ignition buys go | 0.002 [0.0002–0.005] | A stronger price jump that reliably triggers momentum traders (effect up), but aggressive buying is very visible | Gentler and harder to detect, but may fail to trigger the momentum traders (strong effect cost) |
| decoy_rate | Chance per tick of a small trade in the opposite direction | 0.0 [0–0.4] | Blurs the one-directional flow (exec_side_imbalance and long_net_ratio down), but buys back part of what is sold | Purely one-directional flow |
| busy_period_bias | Act mainly when the market is busy | 0.0 [0–1.0] | Abnormal returns and activity are less unusual against a busy background; the schedule drifts while waiting | Acts in quiet markets, where the burst stands out |

In the experiment the strategist relied mostly on free evasion knobs (jitter, groups, rings, busy-period timing and cover trading) and used effect-costing knobs sparingly, which is the behaviour the objective is designed to encourage. The measured trade-offs quoted in the tables (layer spacing, spoof size and cover trading) come from controlled simulations run during development, and they are also given to the strategist in its tactic cards.

## 7.4 CASE-NOTE GENERATOR

When an analyst clicks *Generate case note*, the narrative generator writes a four-to-six sentence explanation of what the account actually did. It does not simply restate the SHAP chart, which the analyst can already see. Instead it receives three inputs: the flag row with its top SHAP features; a short, persona-specific description of the manipulation *mechanism*; and an *evidence* dictionary computed from the account's own orders and trades in the flagged span (`src/agentic/case_evidence.py`). For a layering case, for example, the evidence includes the number of multi-level batches, the median

number of levels per batch, the layer size relative to the market median, the price gap between layers, the share of layer orders cancelled and their median lifetime.

After generation, a grounding check extracts every number in the note and verifies that it appears in the evidence or the flag row. If any number is unsupported, the generator retries once, naming the unsupported numbers; if the note still fails, it is discarded and the dashboard shows only the SHAP evidence. Every note is labelled as AI-generated simulation output.

## 7.5 ANALYST ASSISTANT

The analyst assistant answers free-text questions about the selected run. It has read-only tools that query the run's files: listing rounds, querying round metrics, querying flags with filters, and retrieving an account's history in a round. The assistant is instructed to answer only from tool results and to cite them; the dashboard shows the exact tool calls under a *Sources* expander so the analyst can verify each answer. Context sent to the model is bounded, and the number of tool calls per question is limited; a question that would need more calls is answered with a request to narrow it, rather than a guess.

## 7.6 DASHBOARD

The dashboard (`dashboard/app.py`) is a Streamlit application with a dark theme, a sidebar for selection and four tabs. Figures 7.1 to 7.21 are screenshots of the dashboard showing the experiment analysed in Chapter 8 (run `adversarial_20261014T151727Z_seed42`), mostly at round 5. Every page carries the notice “All data is synthetic. No real trading, real accounts, or real regulatory findings are represented in this application.”

### 7.6.1 SIDEBAR AND NAVIGATION

The sidebar (Figure 7.1) selects the adversarial run, the round (a slider from 0 to the last completed round) and the test market, because each round is evaluated on two independently simulated markets. Below the selectors, a round summary lists precision, recall, F1 and false-positive rate for every persona. The four tabs are reached through a segmented control at the top of the page, and only the selected tab is computed.

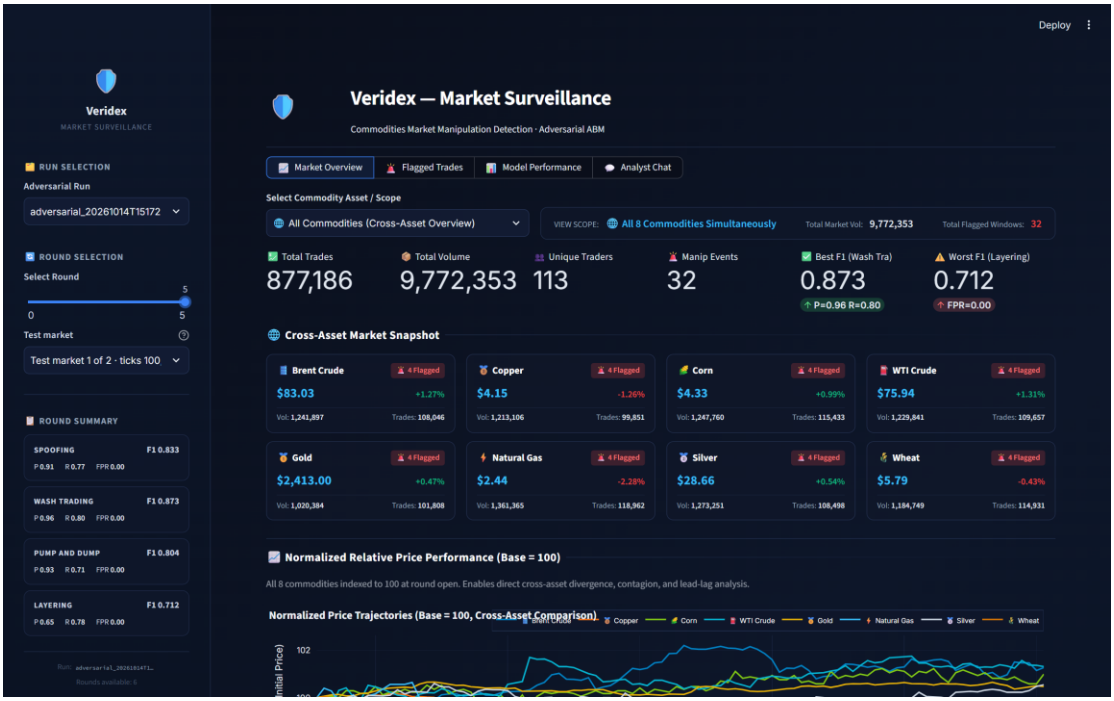



**Figure 7.1: Dashboard with the sidebar (run, round and test-market selection, round summary)**

### 7.6.2 MARKET OVERVIEW

With *All Commodities* selected, the Market Overview tab shows headline figures for the selected market (total trades and volume, unique traders, manipulation episodes, and the best and worst persona F1), a snapshot card for each commodity with its last price, change, volume and number of flagged windows (Figure 7.2), the normalised price paths shown earlier in Figure 5.3, a grid of the eight price profiles with ground-truth manipulation intervals shaded (Figure 7.3) and the correlation matrix (Figure 5.4).

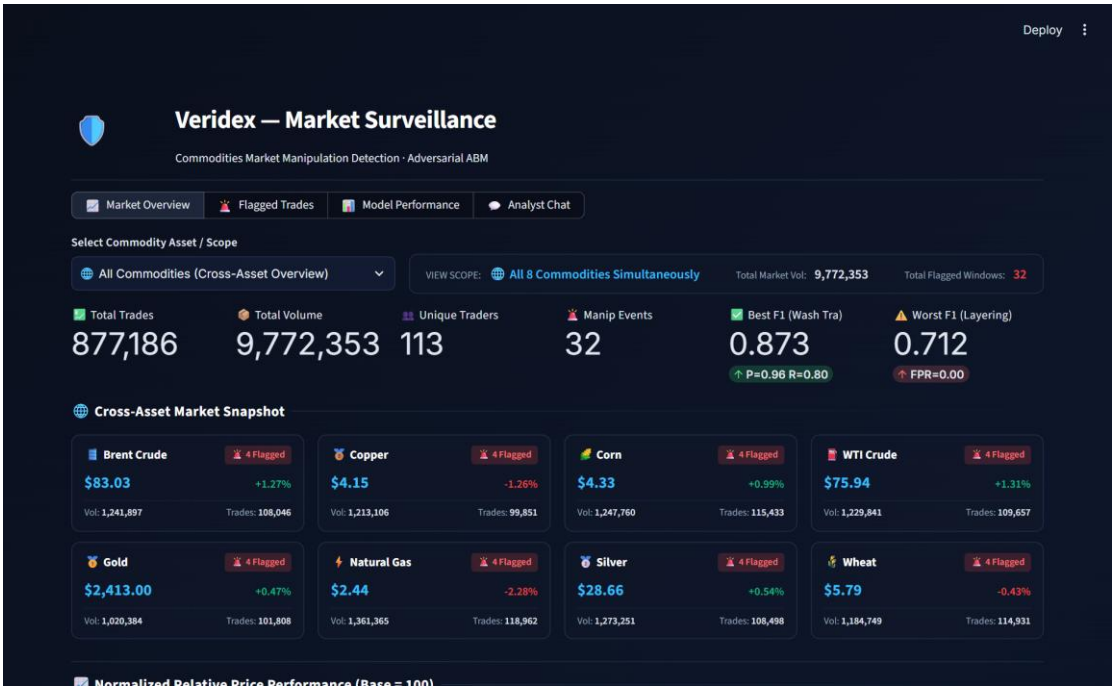



**Figure 7.2: Market Overview: headline metrics and cross-asset snapshot (round 5)**

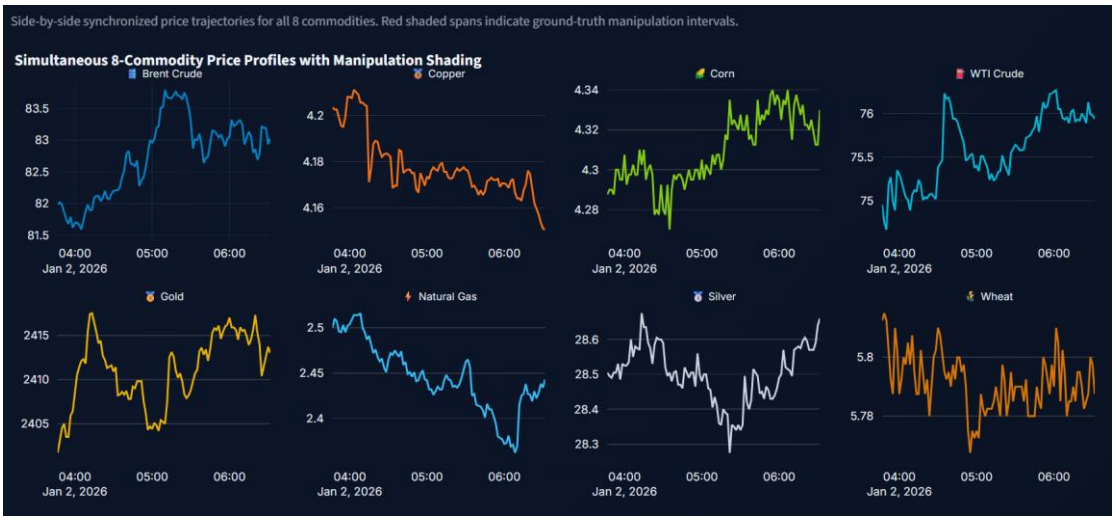



**Figure 7.3: Market Overview: simultaneous price profiles of the eight commodities**

Selecting a single commodity switches to a detailed view (Figure 7.4) that lists the personas active in the market, followed by an OHLCV candlestick chart with each manipulation episode shaded (Figure 7.5), market microstructure series (cancel rate, order-to-trade ratio and buy/sell volume imbalance, Figure 7.6) and a heatmap of the most active accounts over time (Figure 7.7). The heatmap makes the HFT liquidity providers' high activity immediately visible.

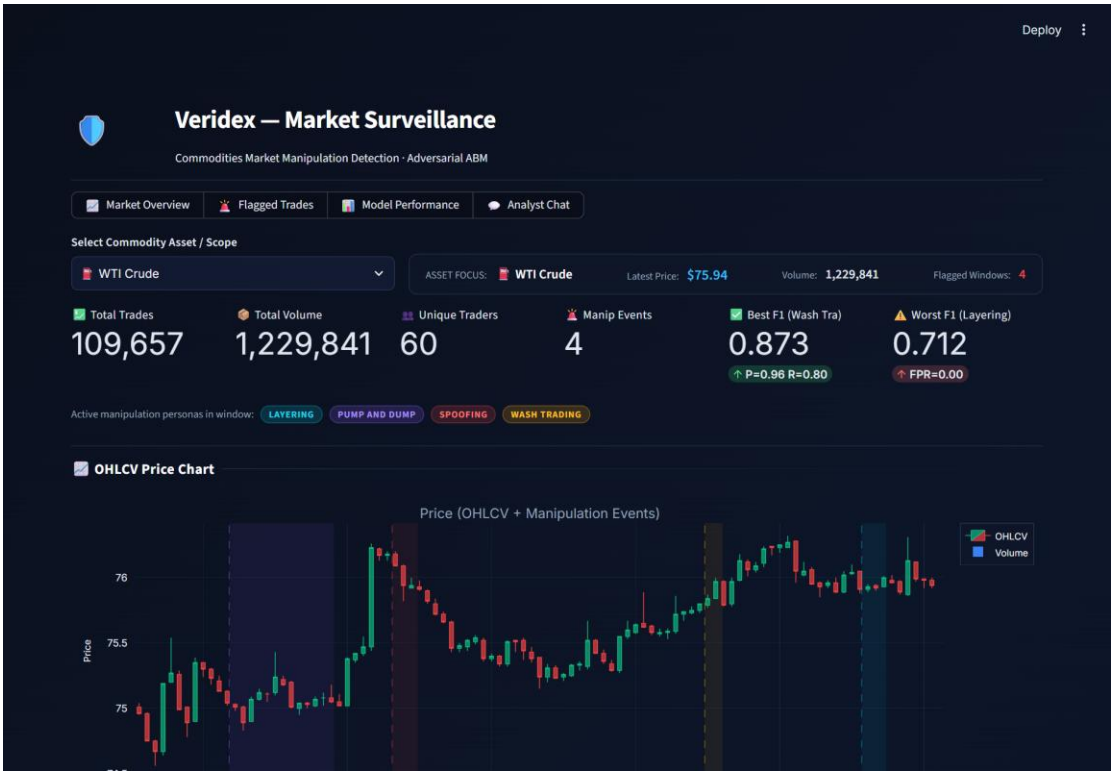



**Figure 7.4: Market Overview for WTI crude oil: headline metrics and active personas**




**Figure 7.5: OHLCV chart of WTI crude oil with manipulation episodes shaded**

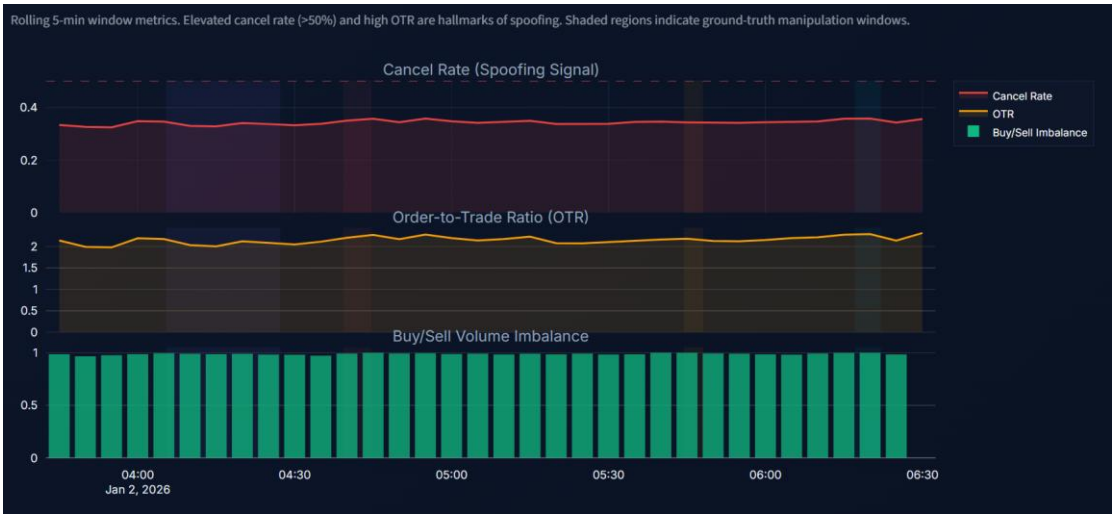



**Figure 7.6: Microstructure series for WTI crude oil**

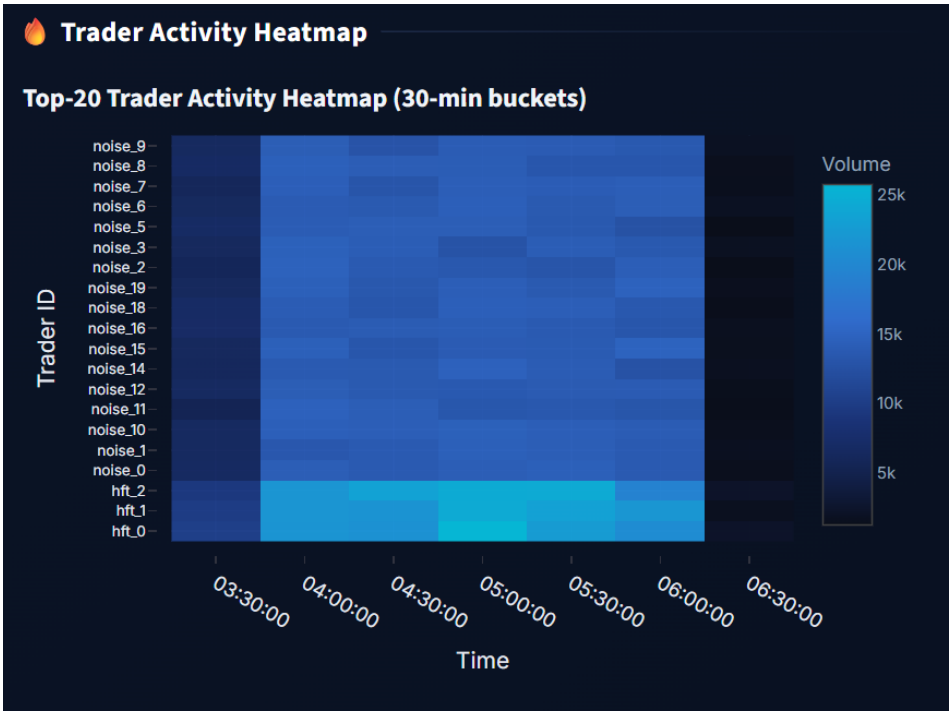



**Figure 7.7: Activity heatmap of the most active WTI accounts (30-minute buckets)**

### 7.6.3 FLAGGED TRADES

The Flagged Trades tab is the analyst's case-review screen (Figure 7.8). Its header counts *persona cases* (consecutive flagged windows of one account and persona merged into a case), how many of them the simulator's ground truth confirms, how many have no matching episode, missed episodes, and windows flagged only by the unsupervised layer. Filters select personas, ground-truth status and a minimum score. Each case is an expandable card (Figure 7.12) containing:

- **Detector belief**: persona, account, commodity, number of windows flagged, score, the layers that fired, and the simulator's ground-truth verdict;
- **Prediction confidence**: the persona classifier's probability against its decision threshold, and the probabilities of all four classifiers, each with its own threshold marker;
- **Price around the case**: the market price with the flagged span shaded and the account's own buys and sells marked;
- **SHAP attribution**: a bar chart of the top five contributions and a table giving each feature's plain-language meaning;
- **What the account did**: the evidence table measured from the account's own orders and trades;
- **Case note**: the AI-generated mechanism explanation with its grounding verdict (Figure 7.10), and the account's raw orders and trades in the span (Figure 7.11).

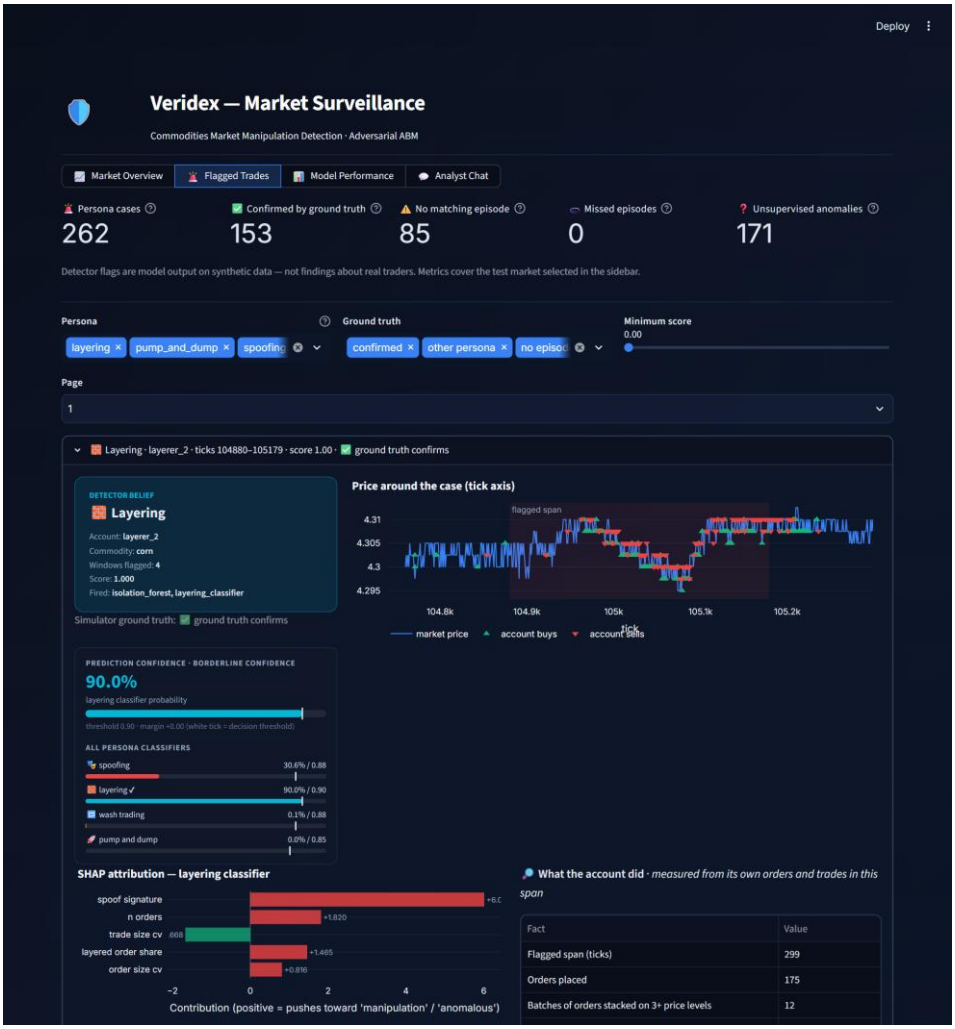



**Figure 7.8: Flagged Trades: case counts, filters and the first case (round 5, test market 1)**




**Figure 7.9: A layering case: detector belief, confidence, price, SHAP and evidence**

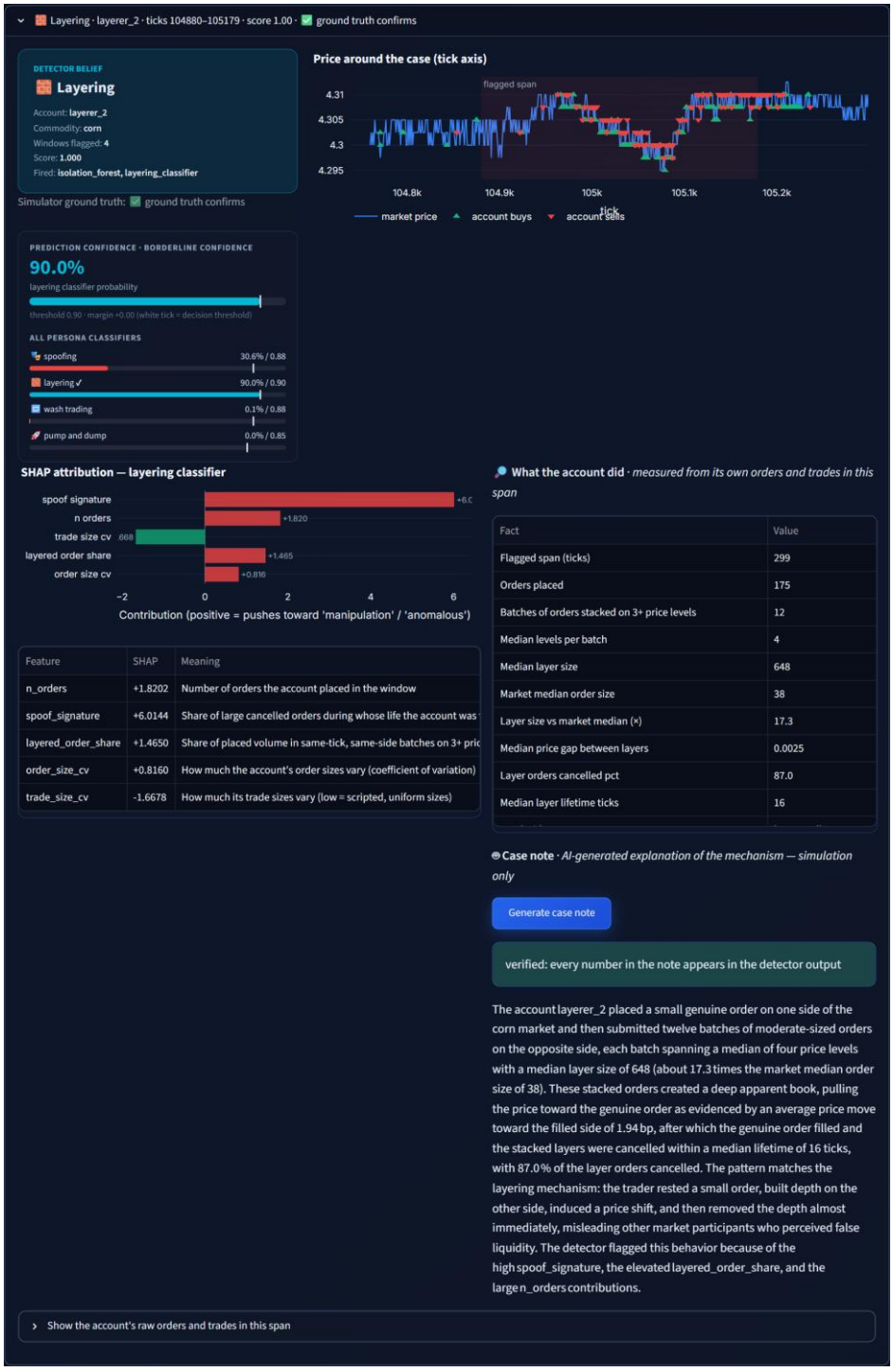



**Figure 7.10: The same case with a verified AI case note**

The case in Figures 7.9 and 7.13 is account `layerer_2` in the corn market. The layering classifier gave a probability of 0.90, just at its threshold of 0.90; the anomaly layer also fired, and the ground truth confirms the episode. The evidence shows 175 orders placed in the 299-tick span, 12 batches stacked on a median of four price levels, layers 17.3 times the market's median order size, 87% of the layer orders cancelled and a median layer lifetime of 16 ticks. The case note explains the mechanism with these numbers,

and the grounding check confirms that every number appears in the detector output. While preparing these screenshots, a defect was found and fixed: the evidence builder had not loaded the order log for layering cases, so the first version of this note reported zero orders, and the raw-order view did not filter by commodity. Both are corrected in the version shown.

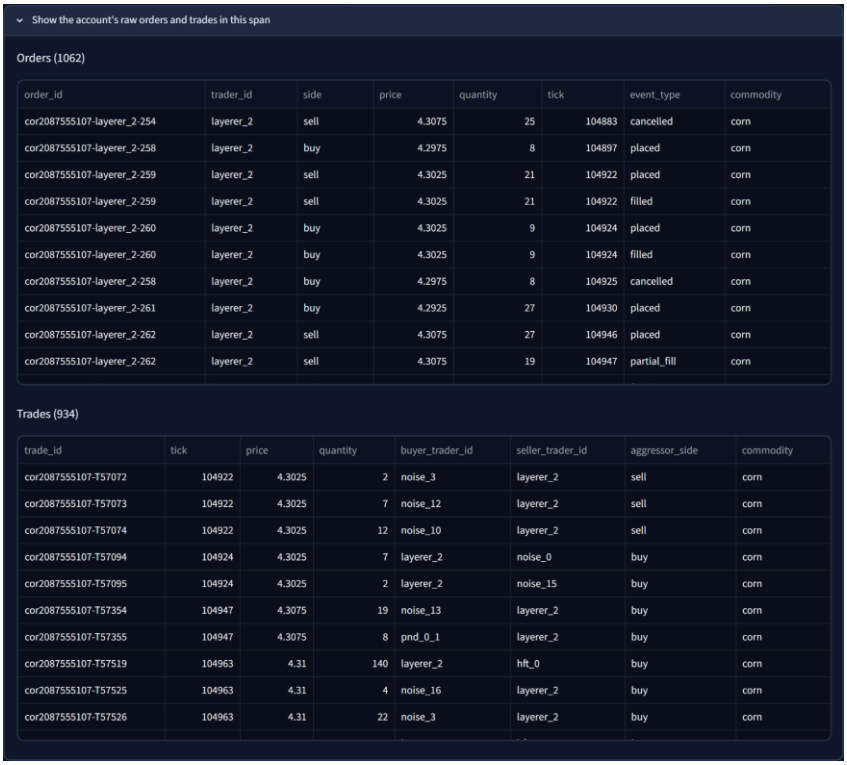



**Figure 7.11: Raw orders and trades of the flagged account in the case span**

### 7.6.4 MODEL PERFORMANCE

The Model Performance tab (Figure 7.12) summarises the latest round (average F1, precision, recall and false-positive rate) and opens with an *Evasion vs. Recovery* panel. A sentence generated from the data highlights the largest drop: in round 5, the spoofing classifier's recall fell to 5% against the frozen detector and recovered to 77% after retraining. The chart beneath it plots recall (or F1, on the second tab) for every persona, with dashed lines for the frozen detector and solid lines after retraining (Figure 7.13).




**Figure 7.12: Model Performance: headline metrics and the Evasion vs. Recovery panel**

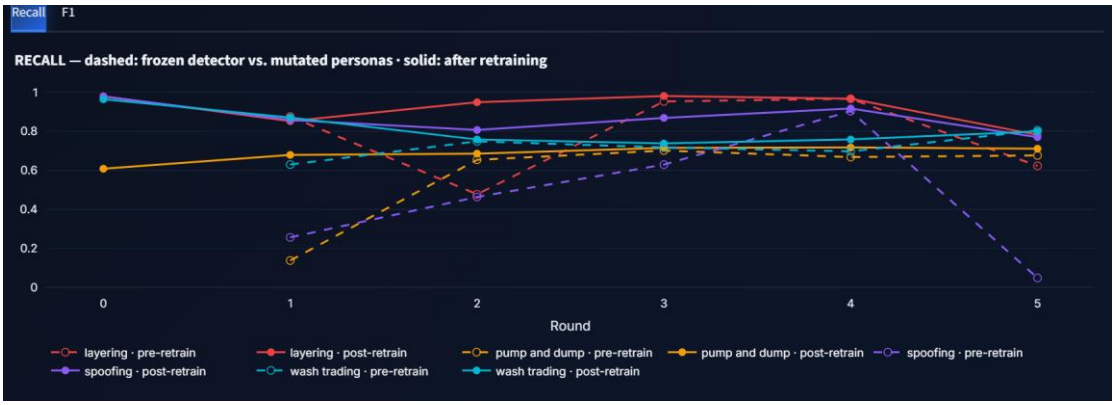



**Figure 7.13: Recall of the frozen (dashed) and retrained (solid) detector by round**

A performance overview follows: an F1 heatmap of persona against round (Figure 7.14) and a radar chart comparing average precision, recall and F1 in round 0 and the latest round (Figure 7.15). The metric-evolution section has tabs for F1, precision, recall and false-positive rate (Figure 7.16), and a precision–recall scatter places every persona-round with marker size proportional to support, against iso-F1 curves (Figure 7.17).




**Figure 7.14: F1 heatmap, persona by round (post-retrain)**

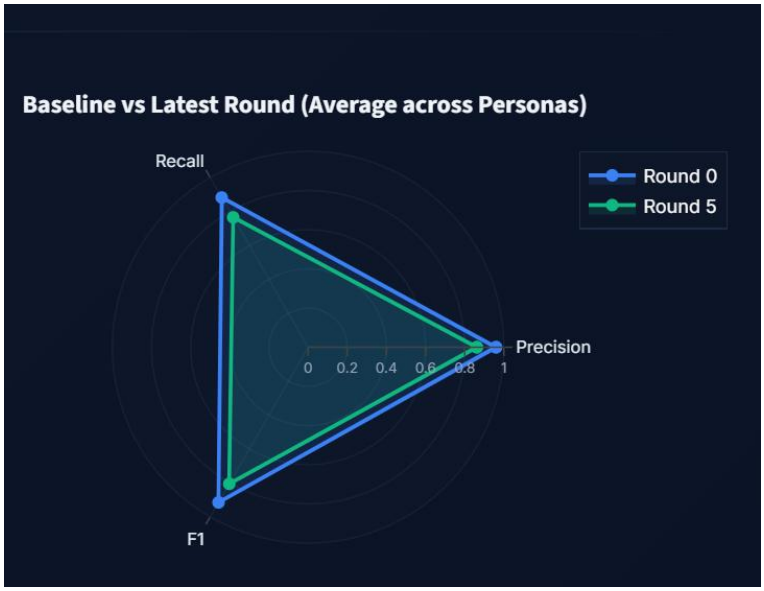



**Figure 7.15: Average precision, recall and F1: round 0 against round 5**

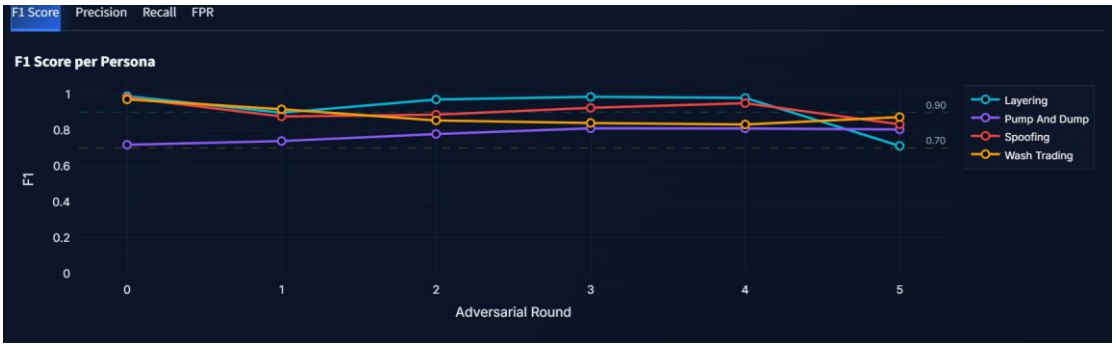



**Figure 7.16: Metric evolution: post-retrain F1 per persona across rounds**

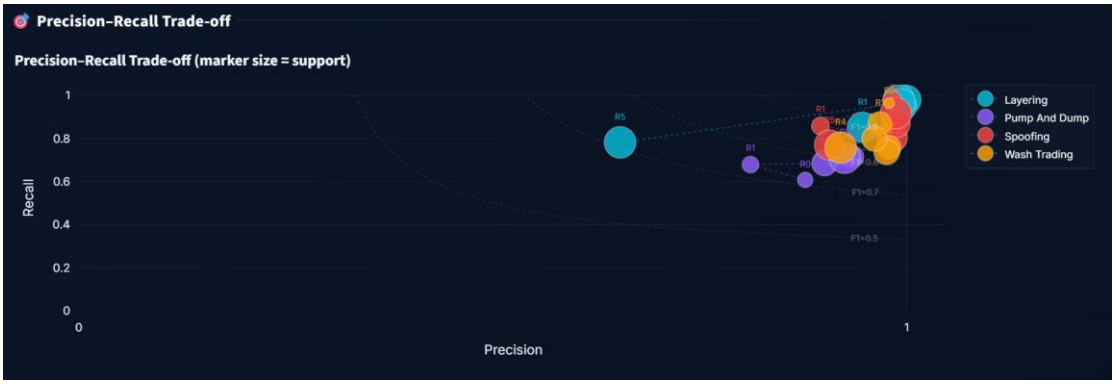



**Figure 7.17: Precision–recall trade-off of every persona and round (marker size = support)**

Two tables complete the quantitative view: the full metrics table, with pre- and post-retrain rows for every persona and round (Figure 7.18), and the parameter-evolution table showing the parameters each persona actually used in each round (Figure 7.19). Finally, the *Adversarial Strategist Decisions* section shows one card per persona and round (Figure 7.20). Each card states whether the LLM or the hand-written fallback chose the parameters, the frozen detector's any-alert coverage and the mutation score, the own-label coverage when it differs, episodes caught and mean delay, retrained recall, effect relative to round 0, the exact parameter changes, and an expandable panel with the strategist's reasoning.

| round | persona | phase | precision | recall | f1 | fpr | support | events_total | events_detected | detection_latency_ticks | episode_coverage | effectiveness_ratio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | Spoofing | post | 0.9826 | 0.9784 | 0.9805 | 0.0000 | 231 | 16 | 16 | 33.5 | 0.9778 | 1 |
| 0 | Wash Trading | post | 0.9792 | 0.9642 | 0.9716 | 0.0001 | 586 | 16 | 16 | 30.6 | 0.9602 | 1 |
| 0 | Pump And Dump | post | 0.8776 | 0.6079 | 0.7183 | 0.0004 | 908 | 16 | 16 | 39 | 0.6102 | 1 |
| 0 | Layering | post | 1.0000 | 0.9796 | 0.9897 | 0.0000 | 196 | 16 | 16 | 32.8 | 0.9809 | 1 |
| 1 | Spoofing | post | 0.8960 | 0.8578 | 0.8765 | 0.0001 | 211 | 16 | 16 | 63.8 | 0.8364 | 0.986 |
| 1 | Wash Trading | post | 0.9676 | 0.8704 | 0.9164 | 0.0004 | 3294 | 16 | 16 | 28 | 0.8628 | 0.97 |
| 1 | Pump And Dump | post | 0.8114 | 0.6789 | 0.7393 | 0.0006 | 1065 | 16 | 16 | 64.1 | 0.68 | 1.139 |
| 1 | Layering | post | 0.9468 | 0.8517 | 0.8967 | 0.0000 | 209 | 16 | 16 | 61.4 | 0.8404 | 0.911 |
| 1 | Spoofing | pre | 1.0000 | 0.2559 | 0.4075 | 0.0000 | 211 | 16 | 13 | 86.5 | 0.2678 | 0.986 |
| 1 | Wash Trading | pre | 0.9933 | 0.6293 | 0.7705 | 0.0001 | 3294 | 16 | 16 | 28 | 0.6152 | 0.97 |

**Figure 7.18: Full metrics table (excerpt)**

| Persona | Parameter | R0 | R1 | R2 | R3 | R4 | R5 |
|---|---|---|---|---|---|---|---|
| Layering | busy_period_bias | 0 | 0 | 0 | 0 | 0 | 0.5 |
| Layering | cancel_jitter | 0 | 0.4 | 0.6 | 0.7 | 0.8 | 0.8 |
| Layering | cancellation_delay_ticks | 5 | 12 | 12 | 12 | 12 | 20 |
| Layering | cover_rate | 0 | 0 | 0.2 | 0.3 | 0.4 | 0.5 |
| Layering | cover_size | 0 | 0 | 0 | 0 | 0 | 0.5 |
| Layering | frequency | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| Layering | layer_size_multiplier | 2 | 2 | 2 | 2 | 2 | 2 |
| Layering | layer_spacing | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 | 0.0002 |
| Layering | n_layers | 4 | 4 | 4 | 4 | 4 | 4 |
| Layering | size_taper | 0 | 0.5 | 0.8 | 0.9 | 0.8 | 0.8 |

**Figure 7.19: Adversarial parameter evolution table (excerpt)**

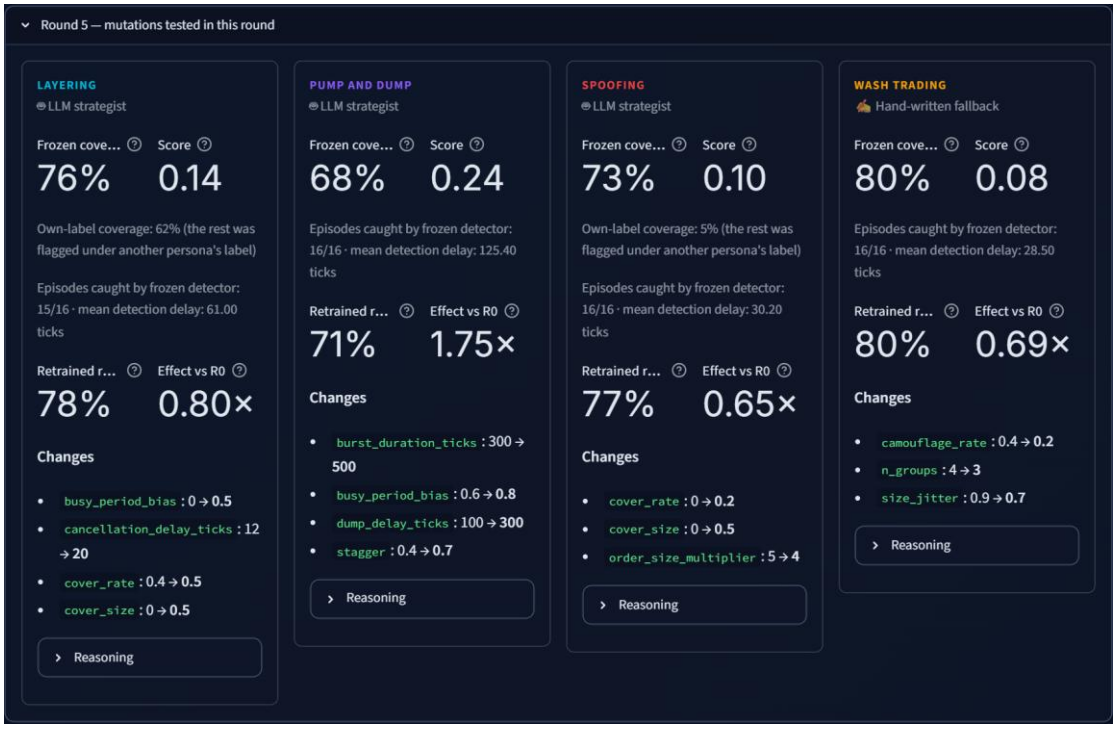



**Figure 7.20: Strategist decision cards for round 5**

### 7.6.5 ANALYST CHAT

The Analyst Chat tab offers suggested questions and a chat box. Figure 7.21 shows the assistant answering "What were the post-retrain precision, recall and F1 for each persona in round 5?". It called `query_round_metrics` for round 5, returned a table whose values match `metrics.json` exactly, and listed the tool call under *Sources*.

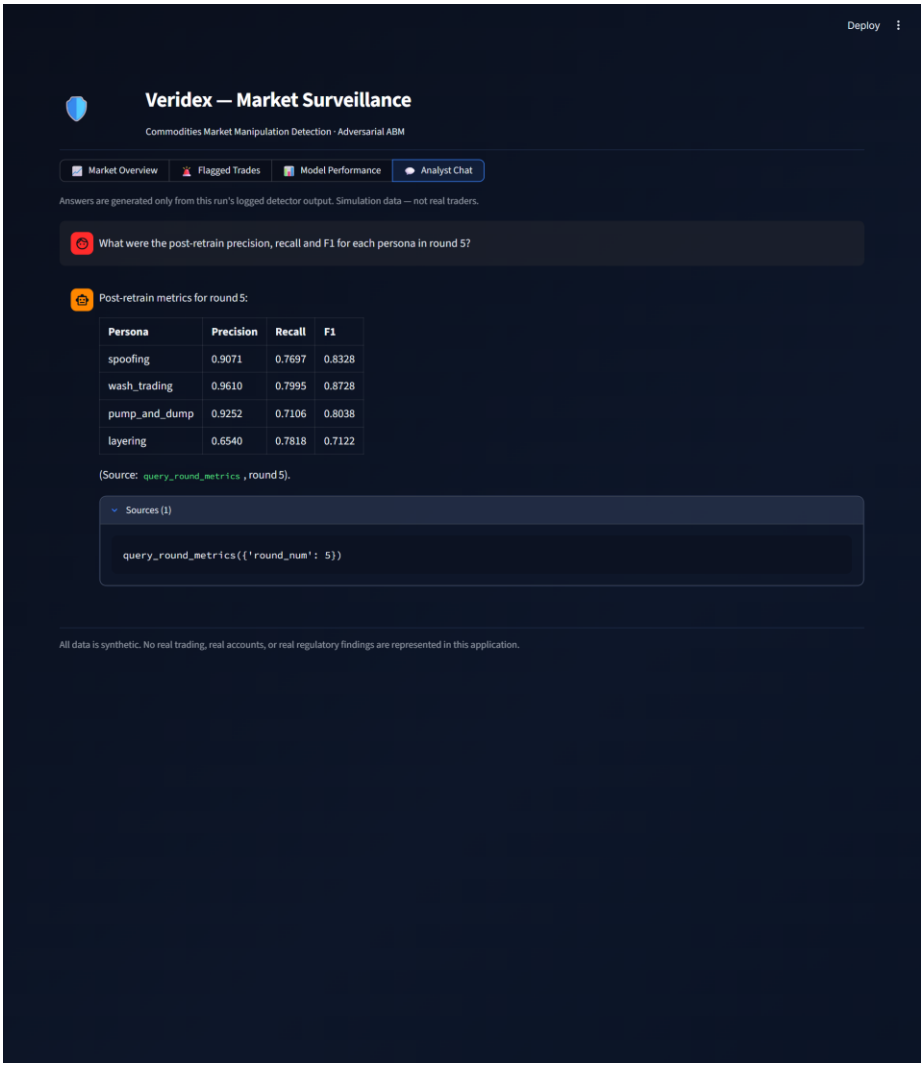



**Figure 7.21: A grounded answer with its sources**

# Chapter 8
# Results and Discussion

This chapter reports the results of a complete six-round adversarial experiment, run `adversarial_20261014T151727Z_seed42`, executed with the final version of the system described in Chapters 5 to 7. All numbers are read directly from the run's output files. Unless stated otherwise, *pre-retrain* (or *frozen*) results score a round's mutated personas with the detector trained before that round, and *post-retrain* results use the detector retrained on all rounds up to and including that round.

## 8.1 EXPERIMENTAL SETUP

**Table 8.1: Experimental setup**

| Item | Value |
|---|---|
| Run identifier / seed | adversarial_20261014T151727Z_seed42 / 42 |
| Rounds | 6 (round 0 baseline, rounds 1–5 adversarial) |
| Markets per round | 1 training market + 2 independent test markets |
| Commodities per market | 8 (WTI, Brent, gold, silver, natural gas, copper, wheat, corn) |
| Length of each market | 10,000 ticks (about 2.8 simulated hours) |
| Legitimate population per commodity | 20 noise, 8 momentum, 6 mean-reversion, 1 market maker, 3 HFT, 3 execution algos |
| Personas | Spoofing, wash trading, layering, pump-and-dump (one episode of each per commodity and market) |
| Episodes evaluated per round | 16 per persona (8 commodities × 2 test markets) |
| Detector | 3-minute windows, 30-second step; trader Isolation Forest (1%); four XGBoost classifiers; OR ensemble |
| Strategist | Cerebras gpt-oss-120b, medium reasoning effort, up to 2,500 output tokens and 5 turns per round |

**Table 8.2: Data volume per round (training and test markets combined)**

| Round | Order events | Trades | Episodes | Account-window rows |
|---|---|---|---|---|
| 0 | 12,401,165 | 2,495,337 | 96 | 326,851 |
| 1 | 12,696,246 | 2,589,925 | 96 | 407,371 |
| 2 | 12,820,864 | 2,636,067 | 96 | 438,061 |
| 3 | 12,763,849 | 2,606,536 | 96 | 421,061 |
| 4 | 12,871,072 | 2,646,646 | 96 | 453,621 |
| 5 | 12,815,646 | 2,629,457 | 96 | 422,161 |

Each round therefore processes about 12.8 million order events and 2.6 million trades. The number of account-window rows grows in later rounds because the strategist added accounts (larger wash rings, more pump-and-dump accounts) and longer episodes.

## 8.2 BASELINE DETECTION (ROUND 0)

**Table 8.3: Round 0 baseline detection (post-retrain, both test markets)**

| Persona | Precision | Recall | F1 | FPR | Episodes detected | Mean latency (ticks) | Coverage |
|---|---|---|---|---|---|---|---|
| Spoofing | 0.983 | 0.978 | 0.981 | 0.0000 | 16/16 | 33.5 | 0.988 |
| Wash trading | 0.979 | 0.964 | 0.972 | 0.0001 | 16/16 | 30.6 | 0.960 |
| Layering | 1.000 | 0.980 | 0.990 | 0.0000 | 16/16 | 32.8 | 0.985 |
| Pump-and-dump | 0.878 | 0.608 | 0.718 | 0.0004 | 16/16 | 39.0 | 0.610 |

Against the obvious, documented versions of the schemes, the detector is strong for the three order-book schemes: F1 of 0.981 for spoofing, 0.972 for wash trading and 0.990 for layering, with every episode detected within about 31–39 ticks (roughly half a minute) of its start and false-positive rates at or below 0.0001. Pump-and-dump is harder (F1 0.718, recall 0.608): its accumulation phase consists of ordinary passive buying that looks like any other trader, so only the ignition and distribution phases are distinctive. On its own, the persona-agnostic trader Isolation Forest reaches precision 0.104 and recall 0.134 against “any manipulation”. This low value is expected once the HFT and execution-algorithm accounts are present: the forest's 1% budget is largely spent on legitimate accounts that are genuinely unusual. Its role in the ensemble is a safety net for schemes the classifiers have not seen.

## 8.3 EVASION: FROZEN-DETECTOR RESULTS

Table 8.4 is the central result of the project. For each mutated round it shows how much of each episode the frozen detector still caught (any-alert coverage, the measure used in the adversary's score), the coverage by the persona's own classifier, the number of episodes caught at all and the mean detection delay.

**Table 8.4: Frozen (pre-retrain) detection of the mutated personas**

| Round | Persona | Any-alert coverage | Own-label coverage | Own-label recall | Episodes caught | Mean delay (ticks) |
|---|---|---|---|---|---|---|
| 1 | Spoofing | 0.955 | 0.268 | 0.256 | 16/16 | 41.2 |
| 1 | Wash trading | 0.619 | 0.615 | 0.629 | 16/16 | 28.0 |
| 1 | Layering | 0.896 | 0.878 | 0.876 | 16/16 | 61.4 |
| 1 | Pump-and-dump | 0.137 | 0.137 | 0.137 | 15/16 | 136.5 |
| 2 | Spoofing | 0.586 | 0.479 | 0.463 | 16/16 | 53.2 |
| 2 | Wash trading | 0.754 | 0.750 | 0.748 | 16/16 | 37.6 |
| 2 | Layering | 0.689 | 0.472 | 0.477 | 15/16 | 98.9 |
| 2 | Pump-and-dump | 0.654 | 0.654 | 0.654 | 16/16 | 51.9 |
| 3 | Spoofing | 0.652 | 0.620 | 0.629 | 16/16 | 51.2 |
| 3 | Wash trading | 0.714 | 0.712 | 0.717 | 16/16 | 35.5 |
| 3 | Layering | 0.952 | 0.952 | 0.953 | 16/16 | 34.6 |
| 3 | Pump-and-dump | 0.702 | 0.702 | 0.701 | 16/16 | 64.0 |
| 4 | Spoofing | 0.912 | 0.900 | 0.903 | 16/16 | 29.3 |
| 4 | Wash trading | 0.699 | 0.697 | 0.697 | 16/16 | 25.8 |
| 4 | Layering | 0.967 | 0.967 | 0.967 | 16/16 | 37.8 |

| Round | Persona | Any-alert coverage | Own-label coverage | Own-label recall | Episodes caught | Mean delay (ticks) |
|---|---|---|---|---|---|---|
| 4 | Pump-and-dump | 0.670 | 0.668 | 0.668 | 16/16 | 74.0 |
| 5 | Spoofing | 0.733 | 0.045 | 0.048 | 16/16 | 30.2 |
| 5 | Wash trading | 0.804 | 0.802 | 0.805 | 16/16 | 28.5 |
| 5 | Layering | 0.758 | 0.616 | 0.623 | 15/16 | 61.0 |
| 5 | Pump-and-dump | 0.679 | 0.677 | 0.676 | 16/16 | 125.4 |

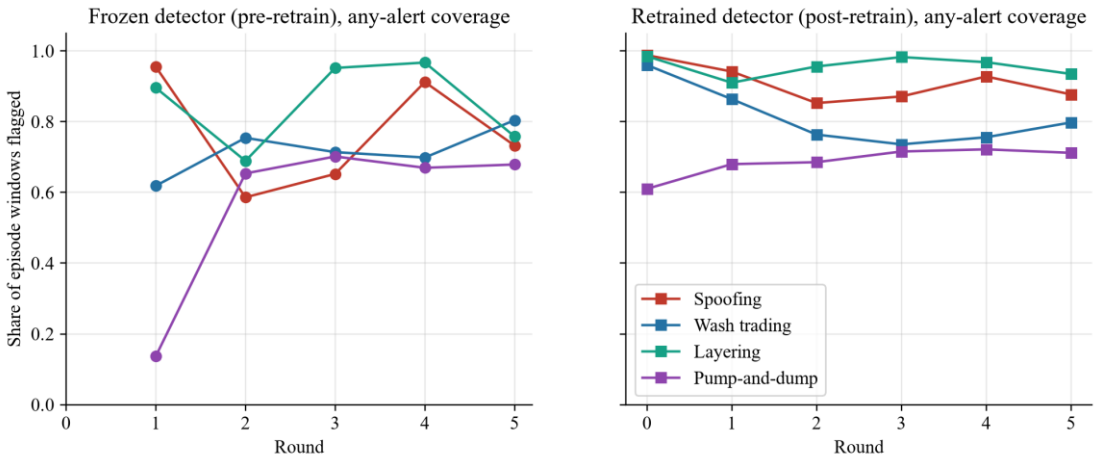



**Figure 8.1: Any-alert coverage of the frozen detector (left) and the retrained detector (right)**

The LLM's mutations clearly challenged the frozen detector for every persona, at different times:

- **Pump-and-dump** was the most successful evasion. In round 1, busy-period timing, staggered account starts and decoy trades reduced the frozen detector's coverage from about 0.61 at baseline to 0.14, missed one of sixteen episodes entirely and delayed detection of the rest to 137 ticks on average. The detector learned the new pattern after retraining, and later mutations held coverage at about 0.65–0.70.
- **Wash trading** fell to 0.62 coverage in round 1 after the strategist split the ring into three groups of four accounts with randomised sizes, a change that dilutes the counterparty-concentration features without reducing the self-traded volume (effect 0.97×). Coverage stayed between 0.70 and 0.80 in later rounds, which makes wash trading the persona the retrained detector recovered least: its retrained coverage also fell from 0.96 to about 0.74–0.80.
- **Spoofing** dropped to 0.59 coverage in round 2 when the strategist introduced cross-account spoofing (genuine and spoof legs on different accounts), smaller spoofs and more size randomisation. In round 5, adding cover trading produced the most striking own-label result of the experiment: the spoofing classifier's

recall fell to 0.048 (own-label coverage 0.045), yet any-alert coverage remained 0.73 because the windows were flagged under the layering label or by the anomaly layer. Scoring the adversary on any alert is what prevents this relabelling from being counted as an escape.

- **Layering** dropped to 0.69 coverage in round 2 (one episode missed entirely, delay 99 ticks) after the strategist tapered layer sizes and started cover trading. In round 5, heavier cover trading lowered the layering classifier's own-label coverage to 0.62 and its precision to 0.32, as many layering windows resembled spoofing and the classifier began flagging other accounts.

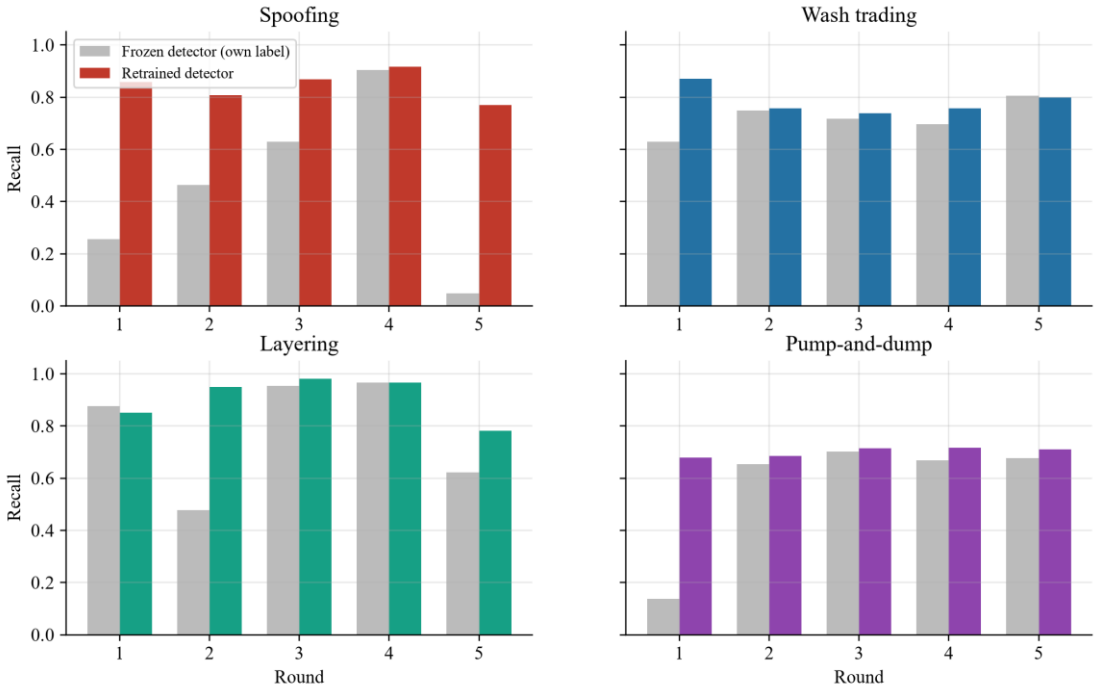



**Figure 8.2: Own-label recall before (grey) and after (colour) retraining, by persona and round**

Figure 8.2 shows the recovery side. In almost every round the retrained detector restores most of the recall the mutation removed, for example spoofing from 0.46 to 0.81 in round 2 and from 0.05 to 0.77 in round 5, and pump-and-dump from 0.14 to 0.68 in round 1. This is the expected behaviour of a regularly retrained supervised detector: a new tactic is dangerous until the detector has seen examples of it.

## 8.4 RECOVERY: POST-RETRAIN PERFORMANCE

**Table 8.5: Post-retrain F1 by persona and round**

| Persona | R0 | R1 | R2 | R3 | R4 | R5 |
|---|---|---|---|---|---|---|
| Spoofing | 0.981 | 0.876 | 0.886 | 0.924 | 0.951 | 0.833 |

| Persona | R0 | R1 | R2 | R3 | R4 | R5 |
|---|---|---|---|---|---|---|
| Wash trading | 0.972 | 0.916 | 0.854 | 0.840 | 0.831 | 0.873 |
| Layering | 0.990 | 0.897 | 0.971 | 0.986 | 0.980 | 0.712 |
| Pump-and-dump | 0.718 | 0.739 | 0.778 | 0.810 | 0.810 | 0.804 |

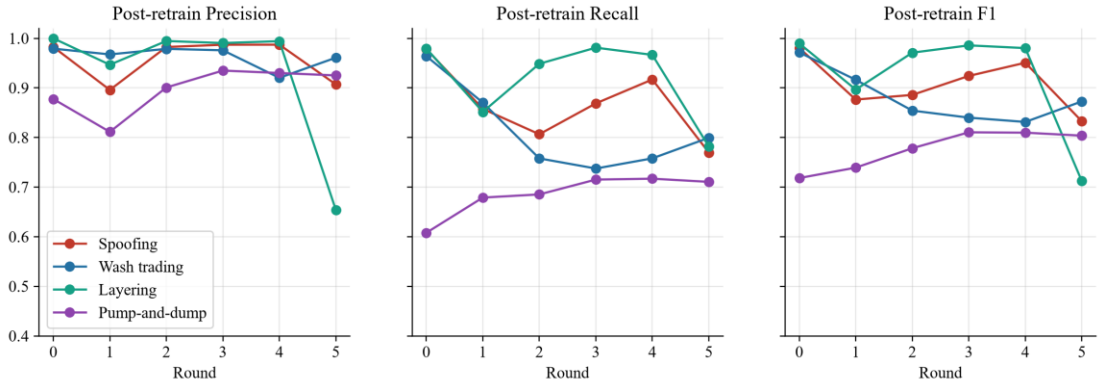



**Figure 8.3: Post-retrain precision, recall and F1 by persona and round**

Retraining keeps precision high throughout (above 0.89 for spoofing and wash trading in every round, and between 0.81 and 0.94 for pump-and-dump), and most of the variation is in recall. Two trends stand out. Wash trading's F1 declines steadily from 0.97 to about 0.83–0.87 because ring and group mutations make its accounts look less like a single pair. Layering stays at 0.90 to 0.99 until round 5, when heavy cover trading brings it to 0.71, mostly through lower precision. Pump-and-dump improves from 0.72 to about 0.80 as the detector accumulates examples of more varied coalitions. Averaged over the four personas, the final round's F1 is 0.805, compared with 0.915 at baseline.

## 8.5 MANIPULATION EFFECTIVENESS AND MUTATION SCORES

**Table 8.6: Effectiveness ratio and mutation score by round**

| Round | Spoof eff. | Wash eff. | Layer eff. | P&D eff. | Spoof score | Wash score | Layer score | P&D score |
|---|---|---|---|---|---|---|---|---|
| 0 | 1.00× | 1.00× | 1.00× | 1.00× | – | – | – | – |
| 1 | 0.99× | 0.97× | 0.91× | 1.14× | 0.05 | 0.20 | 0.08 | 0.54 |
| 2 | 0.83× | 0.99× | 0.82× | 1.70× | 0.20 | 0.15 | 0.20 | 0.21 |
| 3 | 0.62× | 0.72× | 0.80× | 0.96× | 0.13 | 0.12 | 0.04 | 0.18 |
| 4 | 0.74× | 0.71× | 0.80× | 1.93× | 0.05 | 0.12 | 0.03 | 0.21 |
| 5 | 0.65× | 0.69× | 0.80× | 1.75× | 0.10 | 0.08 | 0.14 | 0.24 |

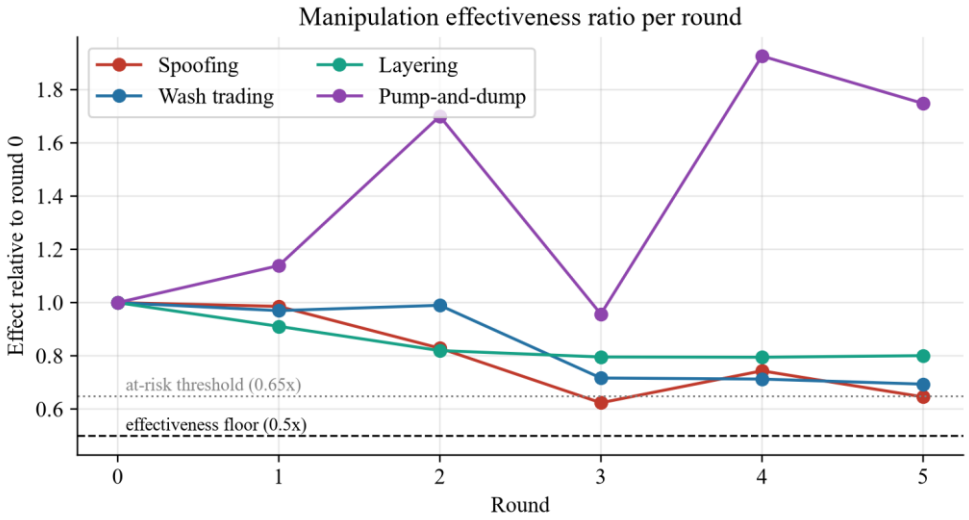



**Figure 8.4: Effectiveness ratio of each persona relative to round 0**

No mutation fell below the 0.5× floor, so every evasion in Table 8.4 was achieved while the scheme was still working. The lowest effect was spoofing in round 3 (0.62×), when the spoof size had been reduced to four times the level depth; the effect guard flagged it as at risk, and the strategist restored part of the size in round 4. Pump-and-dump's ignition pressure rose above the baseline (up to 1.93×) because the strategist spread the coalition's buying over more accounts and longer bursts that happened to concentrate marketable volume; the objective gives no credit for effect above 1.0×, and the surplus note told the strategist so. The highest mutation score of the run, 0.54, was pump-and-dump in round 1 (Figure 8.5).

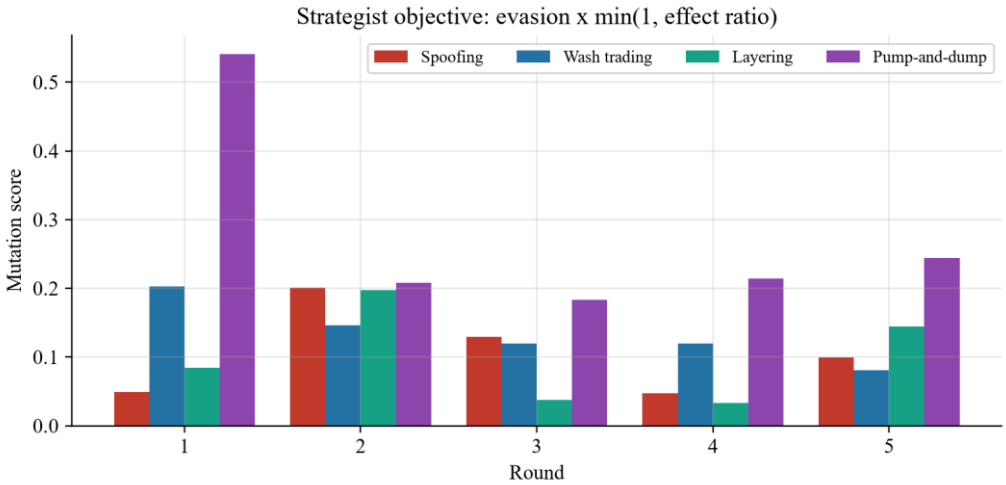



**Figure 8.5: Mutation score (the strategist's objective) per persona and round**

## 8.6 WHAT THE STRATEGIST DID

**Table 8.7: Parameter changes proposed in each round**

| Round | Persona | Source | Changes |
|---|---|---|---|
| 1 | Spoofing | LLM | cancel_jitter 0.0→0.4; cancellation_delay_ticks 5→12; n_split_orders 1→3; size_jitter 0.0→0.4 |
| 1 | Wash trading | LLM | n_groups 1→3; ring_size 2→4; size_jitter 0.0→0.5 |
| 1 | Layering | LLM | cancel_jitter 0.0→0.4; cancellation_delay_ticks 5→12; size_taper 0.0→0.5 |
| 1 | Pump-and-dump | LLM | busy_period_bias 0.0→0.6; decoy_rate 0.0→0.2; stagger 0.0→0.4 |
| 2 | Spoofing | LLM | n_accounts 1→2; order_size_multiplier 8.0→6.0; size_jitter 0.4→0.6 |
| 2 | Wash trading | LLM | busy_period_bias 0.0→0.7; camouflage_rate 0.0→0.2; size_jitter 0.5→0.8 |
| 2 | Layering | LLM | cancel_jitter 0.4→0.6; cover_rate 0.0→0.2; size_taper 0.5→0.8 |
| 2 | Pump-and-dump | LLM | accumulation_size 150→120; busy_period_bias 0.6→0.8; n_coordinated_accounts 3→6 |
| 3 | Spoofing | LLM | n_split_orders 3→4; order_size_multiplier 6.0→4.0; size_jitter 0.6→0.8 |
| 3 | Wash trading | LLM | busy_period_bias 0.7→0.9; camouflage_rate 0.2→0.3; size_jitter 0.8→0.9 |
| 3 | Layering | LLM | cancel_jitter 0.6→0.7; cover_rate 0.2→0.3; size_taper 0.8→0.9 |
| 3 | Pump-and-dump | LLM | burst_duration_ticks 200→300; ignition_aggression 0.002→0.001; n_coordinated_accounts 6→4 |
| 4 | Spoofing | LLM | busy_period_bias 0.0→0.5; cancel_jitter 0.4→0.6; order_size_multiplier 4.0→5.0 |
| 4 | Wash trading | LLM | camouflage_rate 0.3→0.4; n_groups 3→4 |
| 4 | Layering | LLM | cancel_jitter 0.7→0.8; cover_rate 0.3→0.4; size_taper 0.9→0.8 |
| 4 | Pump-and-dump | LLM | busy_period_bias 0.8→0.6; decoy_rate 0.2→0.3; ignition_aggression 0.001→0.0015 |
| 5 | Spoofing | LLM | cover_rate 0.0→0.2; cover_size 0.0→0.5; order_size_multiplier 5.0→4.0 |
| 5 | Wash trading | Fallback | camouflage_rate 0.4→0.2; n_groups 4→3; size_jitter 0.9→0.7 |
| 5 | Layering | LLM | busy_period_bias 0.0→0.5; cancellation_delay_ticks 12→20; cover_rate 0.4→0.5; cover_size 0.0→0.5 |
| 5 | Pump-and-dump | LLM | burst_duration_ticks 300→500; busy_period_bias 0.6→0.8; dump_delay_ticks 100→300; stagger 0.4→0.7 |

Table 8.7 lists every mutation. Nineteen of the twenty persona mutations came from the LLM; in round 5 the model wrote its wash-trading proposal as plain text instead of calling the tool, then reported that all personas had been mutated, and after five turns the system fell back to the hand-written library for that persona. Several patterns show that the strategist used its context rather than changing knobs at random:

- It targeted the features the detector relied on. In round 1, with `order_size_cv` and `fill_ratio` the spoofing classifier's most important features, it increased size jitter, cancel jitter, cancel delay and order splitting, noting that the total spoof depth, and therefore the effect, was unchanged.
- It escalated along documented tactics: cross-account spoofing (round 2), smaller spoofs (round 3), busy-period timing (round 4) and cover trading (round 5), each grounded in a tactic card.
- It reacted to the effect guard: after spoofing's effect fell to 0.62× in round 3, its round-4 reasoning began “order_size_multiplier 4.0->5.0: undo prior down move, restores effect”.

- It adopted the new cover-trading knobs for layering from round 2 and for spoofing in round 5, raising `cover_rate` step by step from 0.2 to 0.5 for layering.

The strategist also showed limitations. It tended to push the same knobs further each round (for example layering's `size_taper` and `cancel_jitter`) rather than revisiting knobs it had never tried, such as `reprice_to_touch` or `flip_after_cancel`, and its stated expectations about which features a change would weaken were sometimes wrong. Figure 8.6 shows the round-2 strategist cards in the dashboard.

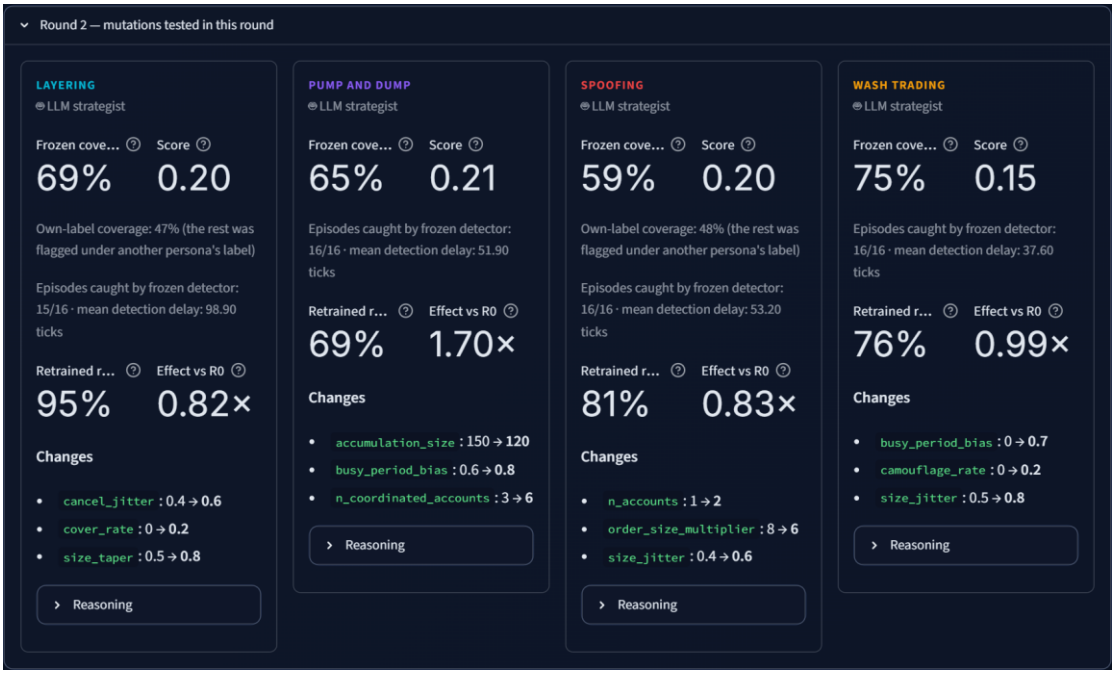



**Figure 8.6: Strategist decision cards for round 2 (dashboard)**

## 8.7 WHAT THE DETECTOR RELIES ON

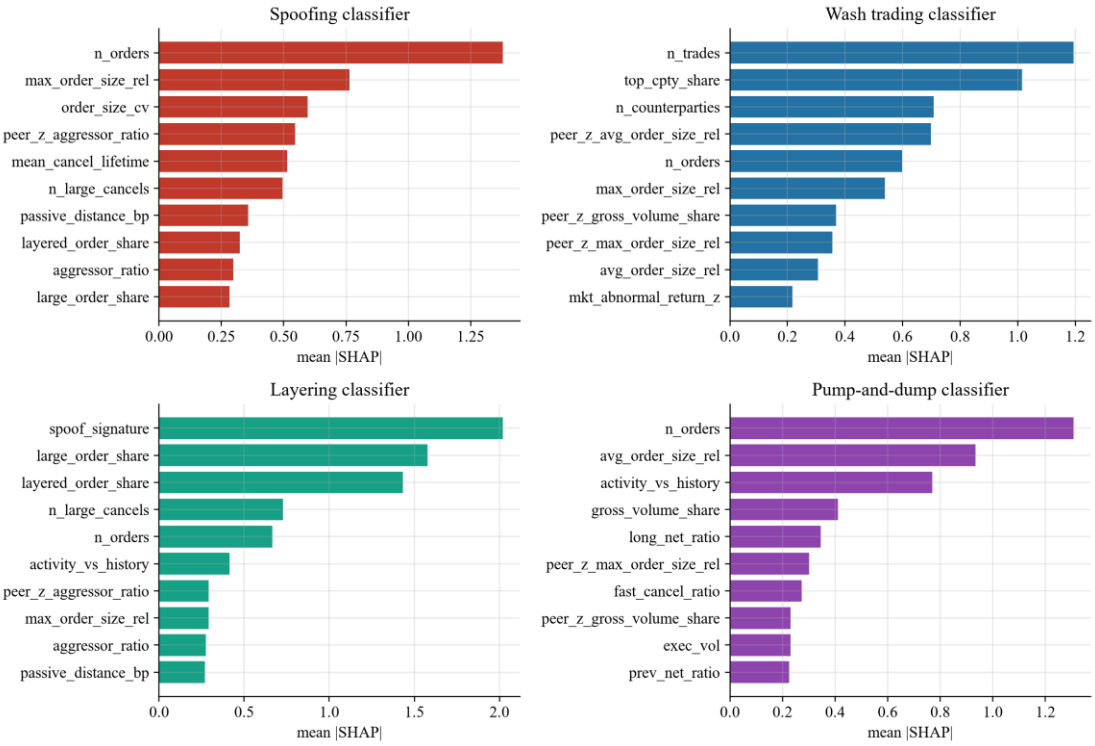



**Figure 8.7: Mean absolute SHAP values of the ten most important features of each classifier, round 5**

Figure 8.7 shows which features each retrained classifier relied on in the final round. The spoofing classifier relies most on `n_orders`, `max_order_size_rel` and `order_size_cv`: after five rounds of size randomisation, splitting and cover trading, the detector has shifted from "large orders cancelled quickly" to "many orders of very irregular size, including some that are still much larger than normal". The layering classifier relies on `spoof_signature`, `large_order_share` and `layered_order_share`, the direct fingerprints of the scheme. The wash-trading classifier relies on `n_trades`, `top_cpty_share` and `n_counterparties`, and the pump-and-dump classifier on `n_orders`, `avg_order_size_rel` and `activity_vs_history`. This drift of reliance between rounds is itself a useful output: it is passed to the strategist, and it shows an analyst which behaviours the detector currently treats as evidence.

## 8.8 FALSE ALERTS ON LEGITIMATE LOOK-ALIKE TRADERS

**Table 8.8: Share of account-windows flagged, by trader type and round (post-retrain detector)**

| Trader type | R0 | R1 | R2 | R3 | R4 | R5 |
|---|---|---|---|---|---|---|
| Manipulation windows (detected share) | 80.1% | 83.3% | 74.8% | 75.1% | 76.7% | 77.4% |
| Execution algorithms | 51.7% | 51.0% | 54.5% | 54.0% | 60.2% | 61.1% |

| Trader type | R0 | R1 | R2 | R3 | R4 | R5 |
|---|---|---|---|---|---|---|
| HFT liquidity providers | 1.2% | 0.5% | 1.3% | 0.7% | 0.9% | 1.4% |
| Noise traders | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% | 0.0% |
| Momentum traders | 0.0% | 0.0% | 0.0% | 0.0% | 0.1% | 0.1% |
| Mean-reversion traders | 0.0% | 0.1% | 0.0% | 0.1% | 0.1% | 0.1% |
| Spoofing accounts outside episodes | 0.3% | 0.4% | 0.1% | 0.1% | 0.4% | 0.4% |
| Layering accounts outside episodes | 0.2% | 0.3% | 0.3% | 0.3% | 0.4% | 1.0% |
| Wash accounts outside episodes | 0.3% | 0.3% | 0.2% | 0.1% | 0.3% | 0.2% |
| Pump-and-dump accounts outside episodes | 0.2% | 0.3% | 0.1% | 0.1% | 0.3% | 0.2% |

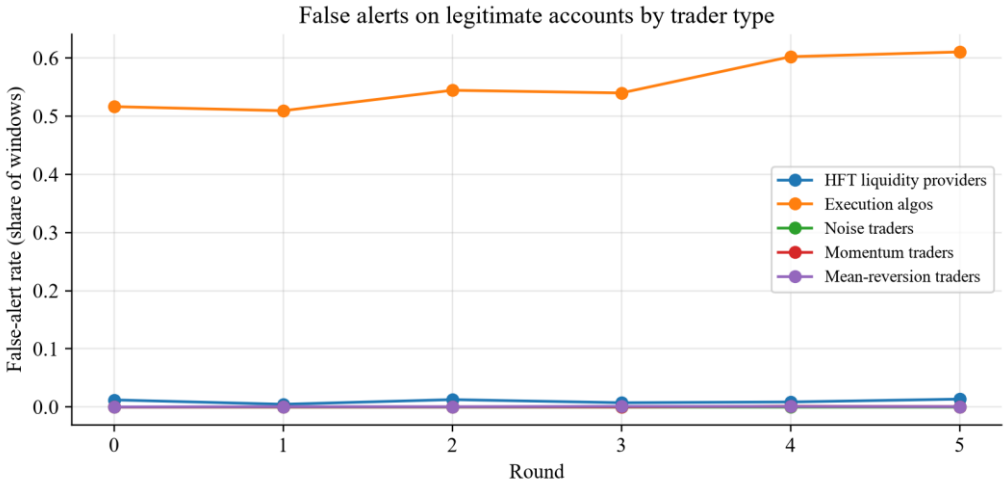



**Figure 8.8: False-alert rate on legitimate accounts by trader type**

Table 8.8 separates the detector's alerts by the type of account that received them. Ordinary participants are almost never flagged: noise traders 0.0%, momentum and mean-reversion traders 0.1% or less, and manipulator accounts outside their episodes below 1%. The HFT liquidity providers, despite cancelling over 90% of their volume, are flagged in only about 0.5–1.4% of their windows. The execution algorithms, however, are flagged in **52–61%** of their windows, and the share grows over the rounds. This is the most important weakness revealed by the experiment. Breaking the round-5 alerts down by the layer that raised them shows the cause: of the 2,386 flagged algorithm windows, 2,383 were raised by the trader Isolation Forest alone and only three by a persona classifier. The algorithms rest and pull orders many times the median size, so they are among the most unusual accounts in every window, and under the OR ensemble every forest flag becomes an alert. In absolute terms the algorithms produce only a few thousand windows per round, compared with over a hundred thousand for noise traders, but in a real surveillance desk this would mean a steady stream of false cases against legitimate institutions. Section 9.2 discusses remedies, such as features that describe whether an account's large orders are ever executed and the relation between its cancellations and its own fills on the other side.

## 8.9 UNSUPERVISED LAYER AND DETECTION DELAY

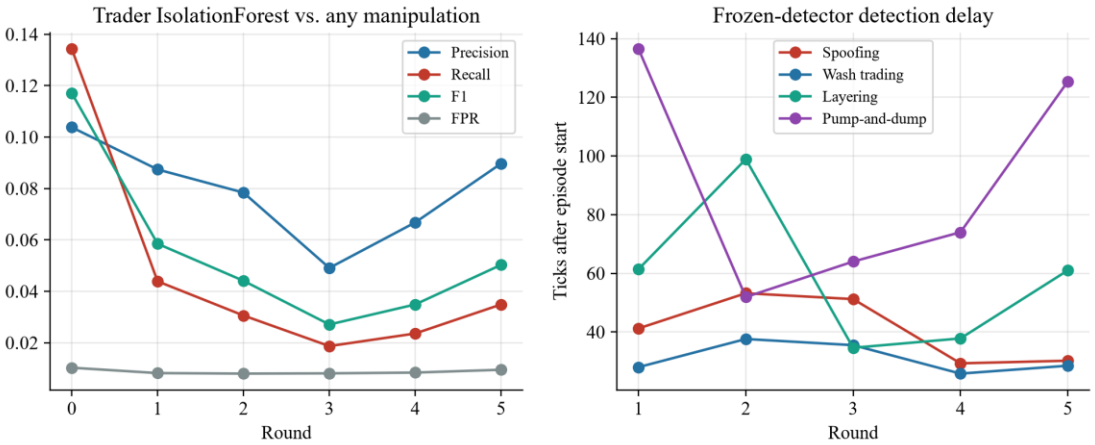



**Figure 8.9: Trader Isolation Forest against any manipulation (left); frozen-detector detection delay (right)**

The left panel of Figure 8.9 shows that the Isolation Forest's recall against all manipulation falls from 0.13 at baseline to about 0.02–0.04 in later rounds, while its false-positive rate stays near its 1% design budget. As manipulators become less extreme and the legitimate look-alikes remain extreme, the most anomalous 1% of accounts is increasingly made up of legitimate firms. The right panel shows the frozen detector's mean detection delay: for most persona-rounds it lies between about 26 and 75 ticks (layering reached 99 ticks in round 2), and successful pump-and-dump mutations stretch it further, to 137 ticks for pump-and-dump in round 1 and 125 ticks in round 5, when longer bursts and slower exits made the scheme's footprint emerge later.

## 8.10 LLM USAGE

The strategist made 22 model calls over the five adversarial rounds and the whole experiment cost about USD 0.07 of API credit. Nineteen of the twenty persona mutations came from the LLM and one from the fallback library (Section 8.6). Answers from the analyst assistant were grounded in tool results, but not always interpreted correctly: in one demonstration it reported post-retrain recall values under the label “frozen detector”. The numbers were real, which is why the dashboard shows every tool call as a source the analyst can check.

## 8.11 SOFTWARE TESTING

**Table 8.9: Automated test suite**

| Suite | What it verifies |
|---|---|
| Order book (unit and property-based) | Price-time priority, partial fills, cancels, no crossed book, quantity conservation under random order streams |
| Agents and personas | Each persona's orders, legs and audit tags; parameter mapping; enabled flags; cross-account spoofing; episode merging |
| Detector | Feature values on hand-built logs, labels, threshold tuning, flags schema, SHAP and persona scores, metrics |
| Effectiveness and objective | Effectiveness measures, objective formula, any-alert scoring, effect guard (reset, at-risk revert, surplus), cover trading |
| LLM layer | Provider switch, budget ledger refusal, cache, error classification, strategist validation and fallback, tool outputs |
| Integration | Byte-identical reproducibility for a fixed seed; a run with no personas enabled |
| Schemas | Every output file validates against its pandera schema |

The suite contains 123 automated tests, all passing on the final code. The tests that involve the LLM use stubbed responses, so the suite never spends API credit.

## 8.12 DISCUSSION

The experiment supports the main hypothesis of the project: a detector that looks excellent on static data (F1 of 0.97–0.99 for three of the four schemes at baseline) can lose a large part of its coverage as soon as an adaptive adversary changes tactics, here down to 0.14 for pump-and-dump, 0.59 for spoofing, 0.62 for wash trading and 0.69 for layering, while the schemes keep at least 62% of their market effect. Evaluating only the retrained detector would have hidden almost all of this, because retraining restores most of the loss within one round. The pre/post-retrain protocol therefore measures two different and complementary properties: robustness to tactics not yet seen, and the ability to learn new ones.

The experiment also shows why the evaluation details matter. Without the effectiveness constraint, the strategist's easiest route would be to shrink the schemes until they stopped manipulating. Without any-alert scoring, relabelling, for example spoofing flagged as layering, would appear as a near-complete escape (round 5 spoofing would have scored own-label coverage 0.045). And without legitimate look-alike traders, the anomaly layer alone catches every spoofer, so no tactic could ever succeed and the detector's apparent strength would be an artefact of an unrealistically easy market. Finally, the high false-alert rate on execution algorithms is a realistic warning: an anomaly layer that is allowed to raise alerts on its own will spend its budget on legitimate but unusual institutions once such institutions are present, so precision on

look-alike populations must be monitored alongside recall. Section 8.13 shows how this was addressed after the experiment.

## 8.13 IMPROVEMENTS MADE AFTER THE EXPERIMENT

The analysis above led to five changes to the system. They are part of the final code but were introduced after the six-round experiment, so the results in Sections 8.1 to 8.12 were produced without them.

- **Self-history features.** Eight new features compare each account with its own earlier, non-overlapping windows (robust z-scores against its own median and inter-quartile range). A legitimate algorithm that always rests large orders looks normal against itself; an account that only now starts spoofing does not.
- **Corroborated anomaly alerts.** An Isolation Forest flag now becomes an alert only if a persona classifier is at least half-way to its threshold or the account deviates strongly from its own history ($|\text{self z}| \geq 6$). Uncorroborated flags are kept as a *review* tier that does not count as an alert and is shown separately in the dashboard.
- **Probe-tested strategist candidates.** The strategist now proposes up to two alternatives per persona, at least one using a knob it has never tried. Every candidate is simulated on the same short probe markets (four commodities, two seeds, 6,000 ticks) and scored against the detector that will be frozen for the next round; an alternative replaces the primary proposal only if its score is higher by at least 0.02, so that small, noisy differences do not decide. The strategist also sees which features caught its own episodes in the previous round, and a proposal written as text instead of a tool call is now accepted.
- **Simulator volatility correction.** A per-commodity factor, fitted by simulation, scales the fundamental volatility so that realised five-minute volatility matches the sourced daily volatility within about 6% for every commodity (previously up to 47% too high).
- **Dashboard fixes.** Layering case evidence now loads the order log (previously a case could show zero orders), and the raw-order view filters by commodity.

The effect of the first two changes was measured in a controlled comparison on the data of the reported experiment: the detector was trained on rounds 0–3 with the new

features and scored on round 4's test markets as a frozen detector, once with the old OR rule and once with corroboration (Table 8.10).

**Table 8.10: Controlled comparison of the alert rule (frozen detector, round 4 of the experiment)**

| Measure | Original detector (from the run) | New features, OR rule | New features, corroborated rule |
|---|---|---|---|
| False-alert rate, execution algorithms | ≈ 60% | 43.9% | **5.8%** |
| False-alert rate, HFT liquidity providers | ≈ 1% | 1.0% | 0.6% |
| False-alert rate, noise traders | 0.0% | 0.0% | 0.0% |
| Any-alert coverage: spoofing | 0.912 | 0.925 | 0.925 |
| Any-alert coverage: wash trading | 0.699 | 0.740 | 0.740 |
| Any-alert coverage: layering | 0.967 | 0.967 | 0.967 |
| Any-alert coverage: pump-and-dump | 0.670 | 0.717 | 0.717 |
| Review-tier windows (not alerts) | – | – | 1,541 |

Corroboration removed about 87% of the false alerts on execution algorithms without losing any manipulation coverage, and the self-history features alone improved coverage for wash trading and pump-and-dump. In a trial round, candidate probing changed the strategist's choice for two of the four personas: for spoofing it selected a cross-account variant with cover trading, and for wash trading a variant with three groups and a lower cross rate, both scoring higher against the frozen detector than the strategist's first proposal. A complete multi-round experiment with all of these changes is the natural next step for confirming their combined effect.

# Chapter 9
# Conclusion and Future Work

## 9.1 CONCLUSION

This project set out to measure how well a market-manipulation detector holds up when manipulators adapt to it, and built Veridex, a complete simulated surveillance laboratory, to do so. Its agent-based simulator produces realistic order books for eight commodity futures with exact manipulation ground truth, and, importantly, contains legitimate high-frequency liquidity providers and execution algorithms whose behaviour overlaps with manipulation. Its two-layer detector combines an Isolation Forest with persona-specific XGBoost classifiers over 51 trader-level features and explains every alert with SHAP values, evidence tables and grounded case notes. Its adversarial loop lets a bounded LLM strategist re-tune the manipulators each round, judges every mutation on both evasion of the frozen detector and remaining market effect, and enforces that judgement in code through any-alert scoring and an effect guard. A Streamlit dashboard makes the whole process inspectable.

The results support the project's central claim. Against the documented baseline schemes, the detector reached F1 scores of 0.97–0.99 for spoofing, wash trading and layering and 0.72 for pump-and-dump. Within one or two rounds, the LLM strategist found mutations that reduced the frozen detector's coverage to 0.14 for pump-and-dump, 0.59 for spoofing, 0.62 for wash trading and 0.69 for layering while keeping at least 62% of each scheme's effect, using documented tactics such as cross-account spoofing, staggered coalitions, ring splitting and cover trading. Cumulative retraining recovered most of each loss, and the final round's post-retrain F1 averaged 0.805. The whole experiment cost about USD 0.07 of API credit.

The project also produced several lessons that were not obvious at the start. An adversary must be required to keep manipulating, or its best strategy is to stop. It must be scored on whether it was caught at all, not on whether it was caught under the right label. A simulator without realistic look-alike traders makes detection artificially easy and adversarial evaluation meaningless. And an anomaly layer that may raise alerts on its own spends its budget on legitimate but unusual institutions: in this experiment 52–61% of execution-algorithm windows were flagged, almost all by the Isolation Forest alone. Requiring corroboration, together with features that compare each account with

its own history, reduced that rate to 5.8% without losing any manipulation coverage (Section 8.13). Each of these lessons is now built into the system or measured by it.

## 9.2 LIMITATIONS

- **Single experiment.** The results come from one six-round run with one seed. Effects of the size reported here are large, but confidence intervals would require repeated runs with different seeds.
- **False alerts on execution algorithms.** In the reported experiment the detector flagged a large share of legitimate execution-algorithm windows. The post-experiment fix (Section 8.13) lowers this to about 6%, which is still higher than for any other legitimate group.
- **Simulator fidelity.** In the reported experiment, realised volatility was up to about 1.5 times the target for copper and silver (the post-experiment correction brings every commodity within about 6% at five minutes). The experiments used long-run published volatilities rather than the higher 2021–2026 yfinance estimates, and the market has no cross-commodity linkages, news events or intraday seasonality.
- **Scope of personas.** Cross-market spoofing, cornering, banging the close and collusion across personas are not modelled, and each episode uses one persona at a time.
- **LLM behaviour.** In the experiment the strategist sometimes repeated the same knobs instead of exploring and once wrote a proposal as text instead of calling its tool (both addressed afterwards by probe-tested alternatives and text parsing), and the analyst assistant once reported correct numbers under the wrong label.
- **Requirements not implemented.** Counterfactual replay and the "price impact avoided by early detection" measure (SRS FR-1.9 and FR-3.5) were not built.

## 9.3 FUTURE WORK

- **Reduce the remaining false alerts on legitimate large traders** with features that separate execution from deception: whether an account's large orders are ever filled, the timing relation between its cancels and its own opposite-side

fills, and consistency of direction over the parent-order horizon. Monitoring precision on look-alike populations as a first-class metric would make this trade-off visible every round.

- **Repeated experiments and statistics**: run each configuration with several seeds and report confidence intervals, and compare the LLM strategist with the hand-authored library and with random or Bayesian search over the same knobs.
- **A co-evolving defender**: let a second agent propose detector changes (new features, thresholds or retraining schedules) in response to the strategist, turning the loop into a two-player game.
- **Market-specific calibration**: drive the simulator directly with the yfinance-fitted volatilities (or with a rolling recent window) to study manipulation in turbulent regimes such as the 2022 energy shock.
- **Richer markets and schemes**: correlated commodities and cross-market spoofing, news-driven volatility, opening and closing auctions (for banging the close), and multi-persona collusion.
- **Counterfactual impact**: replay each market from the detection tick without the manipulator to estimate the price impact that early detection would avoid.
- **Better LLM grounding**: give the analyst assistant explicit pre- and post-retrain query parameters, and validate that labels as well as numbers in its answers match the tool output.
- **Validation on real data**: test whether features and models trained in simulation transfer to anonymised exchange data or published enforcement cases.

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# APPENDICES

## APPENDIX 1 RUNNING VERIDEX

All commands are run from the repository root on Windows PowerShell, using the project's virtual environment.

```
# 1. Create the environment and install dependencies
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt

# 2. Configure the LLM provider (the key is never committed)
#    .env:  LLM_PROVIDER=cerebras
#           CEREBRAS_API_KEY=<your key>

# 3. Run the full adversarial experiment used in Chapter 8
.\.venv\Scripts\python.exe scripts/run_full_pipeline.py --rounds 6 --ticks 10000 --
seed 42 --commodity all

#    Optional: hand-authored mutations only (no API calls)
.\.venv\Scripts\python.exe scripts/run_full_pipeline.py --rounds 6 --mutations hand

# 4. Open the dashboard
.\.venv\Scripts\python.exe -m streamlit run dashboard/app.py

# 5. Run the automated tests
.\.venv\Scripts\python.exe -m pytest tests -q
```

## APPENDIX 2 REPOSITORY STRUCTURE

```
Veridex/
  configs/            detector_config.yaml, llm_config.yaml, persona_config.yaml
  dashboard/          app.py, data_loader.py, styles.py, components/ (sidebar, market, flags,
metrics, chat)
  docs/               CHANGES_overhaul.md, enforcement_case_notes.md, tactic_cards.yaml
  proj_details/       SRS.md, Implementation_Plan.md, Commodities_ABM_Project_Plan.md
  scripts/            run_simulation.py, run_full_pipeline.py
  src/
    core/             order_book.py, simulation.py, calibration.py, persona_params.py
      agents/         base, noise, momentum, mean_reversion, market_maker,
hft_liquidity_provider, execution_algo
        manipulation/ common, spoofing, layering, wash_trading, pump_and_dump
    detector/         features.py, anomaly.py, classifiers.py, pipeline.py
    adversarial/      orchestrator.py, metrics.py, effectiveness.py
    agentic/          llm_client.py, adversarial_strategist.py, tools.py, mutation_library.py,
                      narrative_generator.py, case_evidence.py, analyst_assistant.py
    utils/            schemas.py, io.py, logging_config.py
  tests/              unit/, integration/, property/
  data/runs/<run_id>/ round_N/{train,test,test_1}/, metrics.json, flags.parquet,
strategist.json, llm_calls.jsonl
```